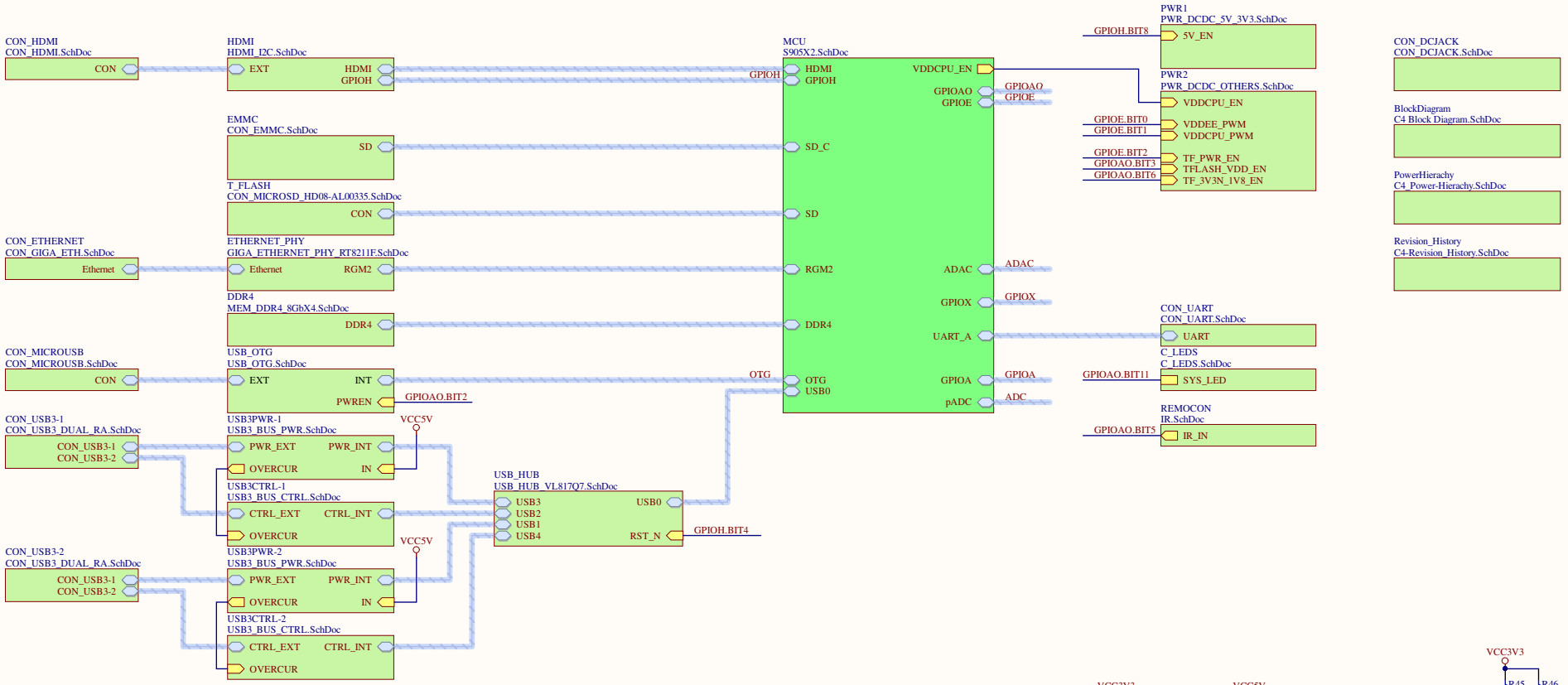
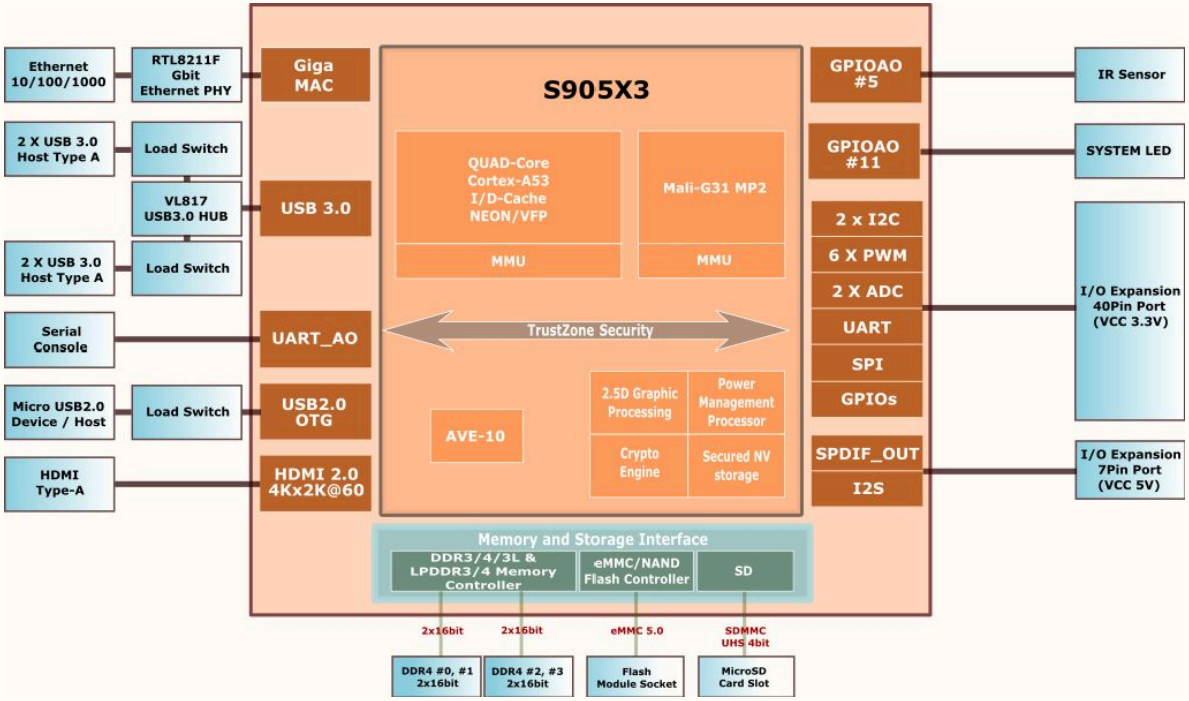


Title	C4 MAIN		Hardkernel Co.,LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Size: A3	Number: 1	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:10 PM	Sheet 1 of 33	
File: C4-MAIN.SchDoc		Designed by: neal.kim@hardkernel.com	

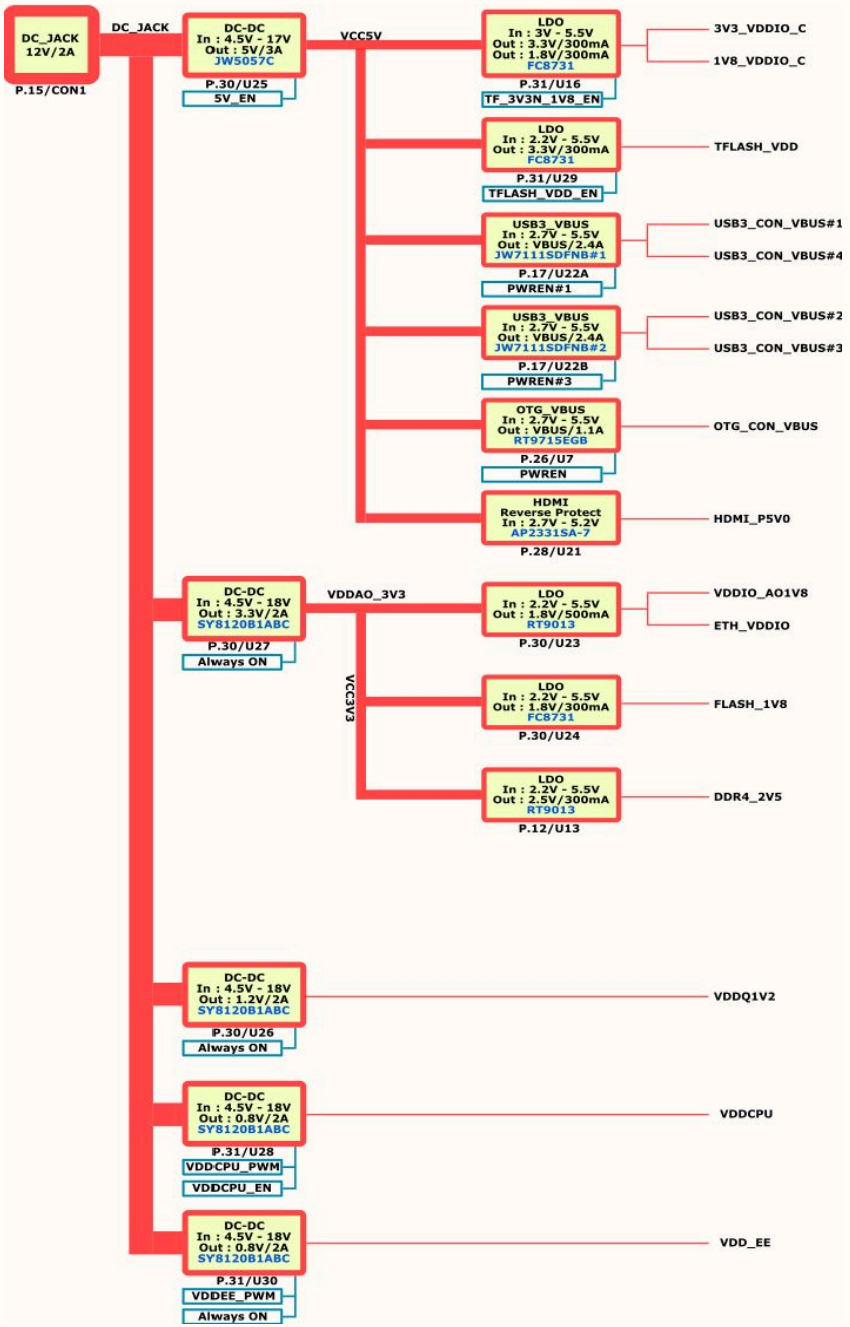
ODROID



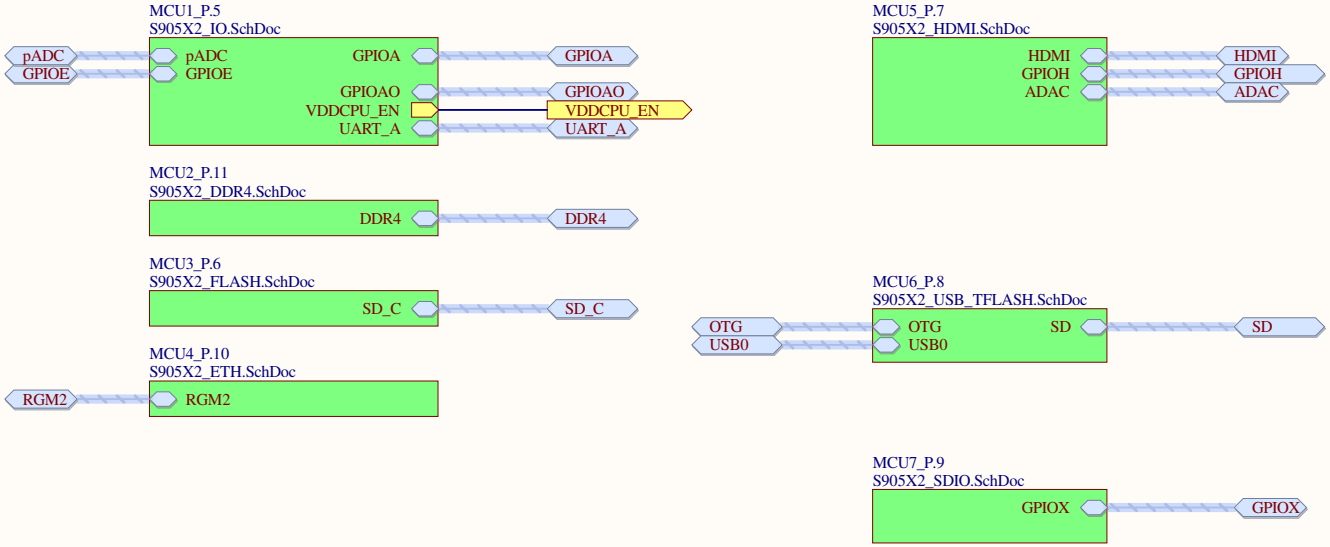
Title: C4 Block Diagram			Hardkernel Co.,LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Size: A4	Number: 2	Revision: 0.4	
Date: 2/17/2020 Time: 2:23:10 PM	Sheet 2 of 33		
File: C4 Block Diagram.SchDoc		Designed by: ruppi kim@hardkernel.com	

HW REVISION	DATE	Change Note
v1.0	2020.01.29	1. Added silkscreen certification mark.
v0.4	2019.11.29	1. Change USB_OTG Power Enable pin from GPIOH.BIT6 to GPIOAO.BIT2 2. Remove eMMC_nDET pin 3. Change TFLASH_VDD Enable pin from GPIOAO.BIT8 to GPIOAO.BIT3 4. Change TF_3V3N_1V8 Enable pin from GPIOAO.BIT7 to GPIOAO.BIT6 5. Asign GPIOE.BIT2 to the TF Power to turn ON/OFF 6. Rearrange Expansion Port 40Pin and 7Pin. 7. Change USB Load S/W from MP62551 to JW7111S 8. Change 5V DCDC Step-down from SY8120B1ABC to JW5057C 9. Add two spare land 3225 size for Cap 220uF next to USB 3.0 Connector
v0.3	2019.09.24	1. Removed the components related to CVBS OUT of S905X3 2. Changed R104 to 39.2k from 16k and R103 to 301k from 120k 3. Changed R108 to 56k from 10k 4. Changed PCB thickness from 1.2T to 1T and layer from 8 to 6. 5. Added L13 and C81 between AVDD0V8_HDMI of S905X3 and VDD_EE 6. Changed net and added components at VBUSDET of VL817.
v0.2	2019.07.19	1. Changed USB 3.0 HUB IC from VL817 to VL805. 2. Changed the connection of PIN35 from ADAC.LOLP to GPIOE.BIT2 on 40Pin Header. 3. Changed the connection of PIN36 from ADAC.LORP to GPIOH.BIT5 on 40Pin Header. 4. Added 2pin Header for Audio out (ADAC.LOLP, ADAC.LORP) 5. Removed external CPU RESET IC with the components. 6. Added 2pin Header for external 12V power input. 7. Changed PCB thickness from 1T to 1.2T and layer from 6 to 8.
v0.1	2019.05.20	1. The first revision.

Title: Revision History			Hardkernel Co.,LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Size: A3	Number: 3	Revision: 0.4	ODROID
Date: 2/17/2020 Time: 2:23:10 PM	Sheet 3 of 33		
File: C4-Revision_History.SchDoc			Designed by: neal.kim@hardkernel.com

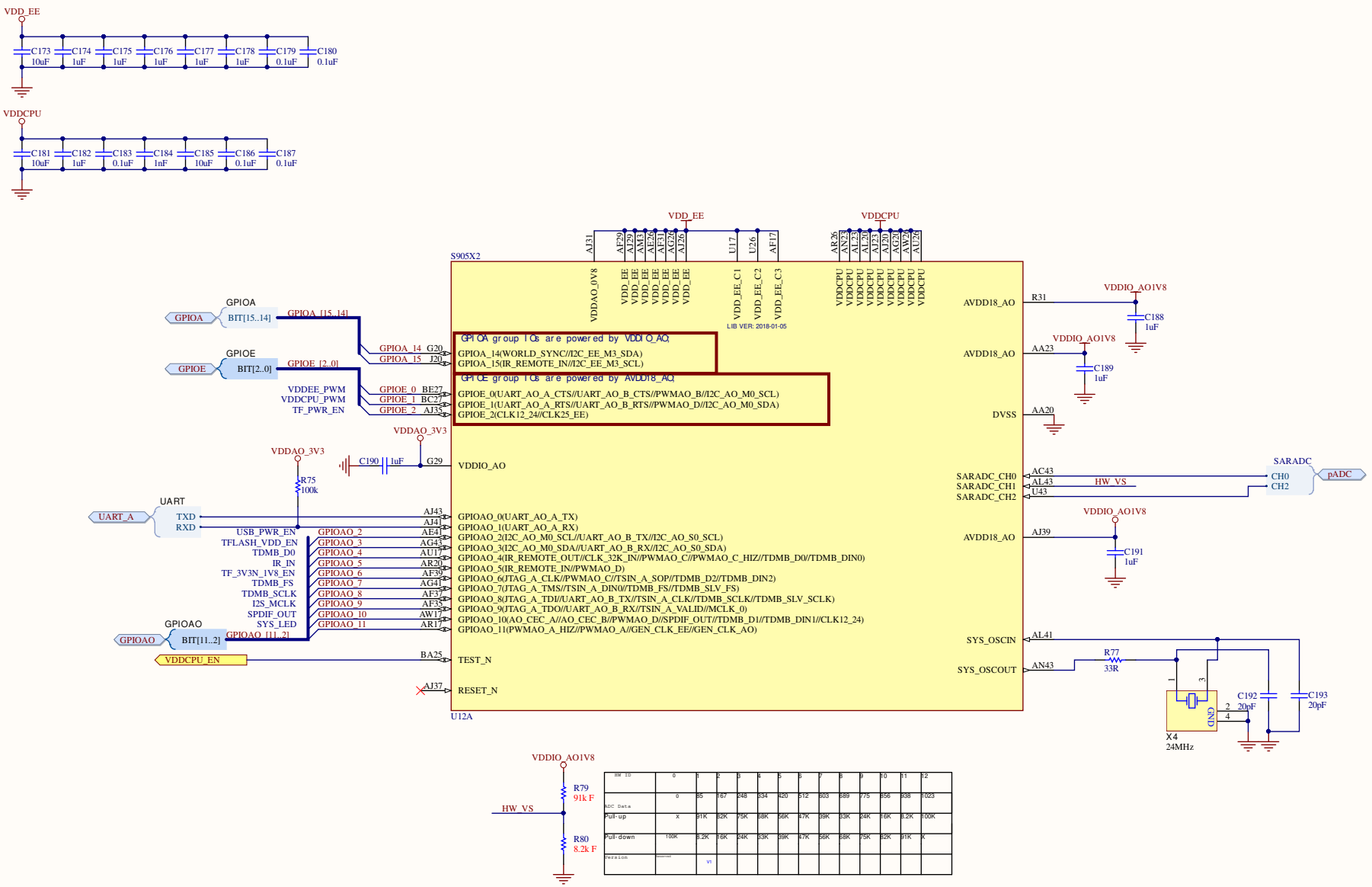


Title: Power Hierachy			Hardkernel Co.,LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Size: A3	Number: 4	Revision: 0.4	ODROID
Date: 2/17/2020	Time: 2:23:10 PM	Sheet 4 of 33	
File: C4_Power-Hierachy.SchDoc		Designed by: ruppi.kim@hardkernel.com	

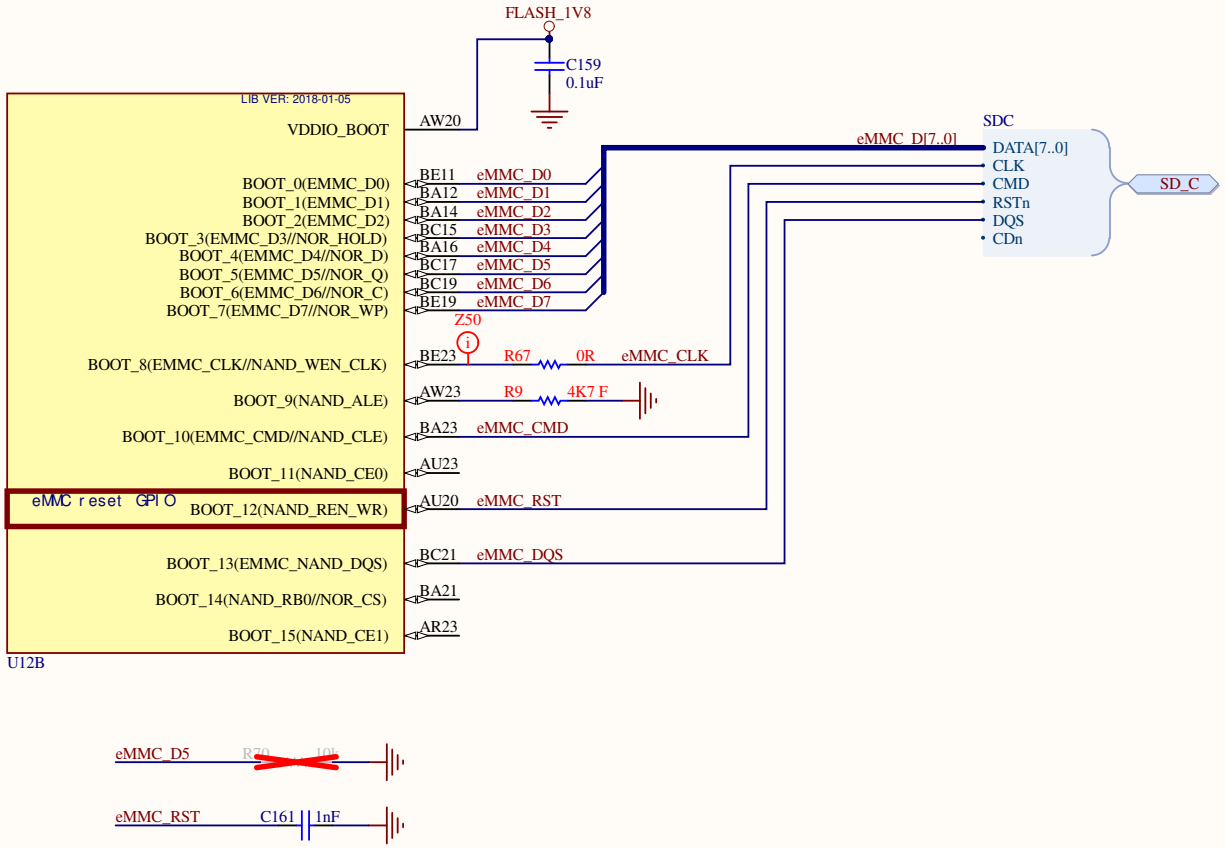


Title S905X2			Hardkernel Co.,LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Size: A4	Number: 5	Revision: 0.4	
Date: 2/17/2020 Time: 2:23:10 PM	Sheet 5 of 33		
File: S905X2.SchDoc			Designed by: ruppi kim@hardkernel.com

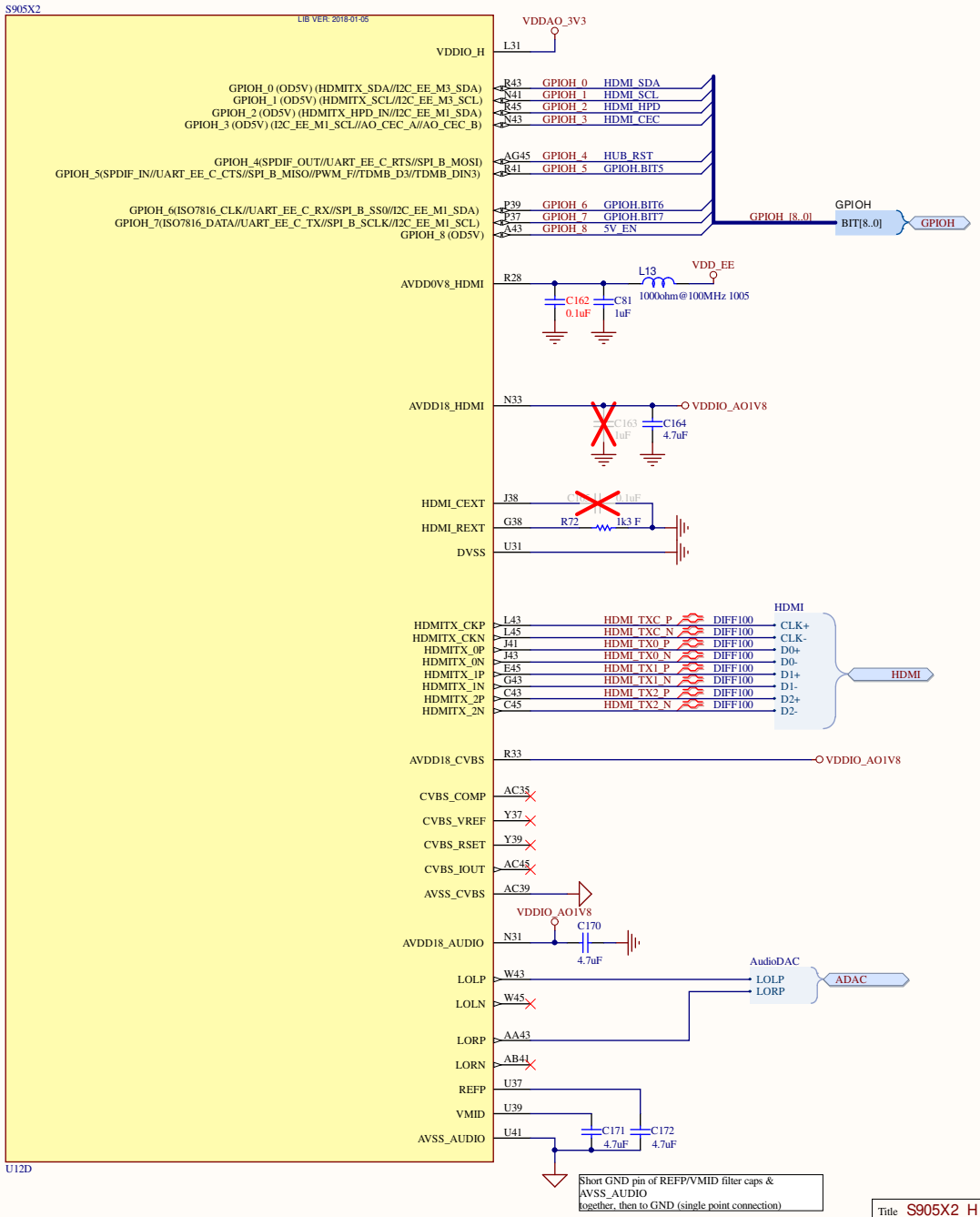
ODROID



Title	S905X2 IO		
Size: A3	Number: 6	Revision: 0.4	Hardkernel Co., LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Date: 2/17/2020 Time: 2:23:10 PM	Sheet 6 of 33		ODROID
File: S905X2_IO.SchDoc			Designed by: neal.kim@hardkernel.com

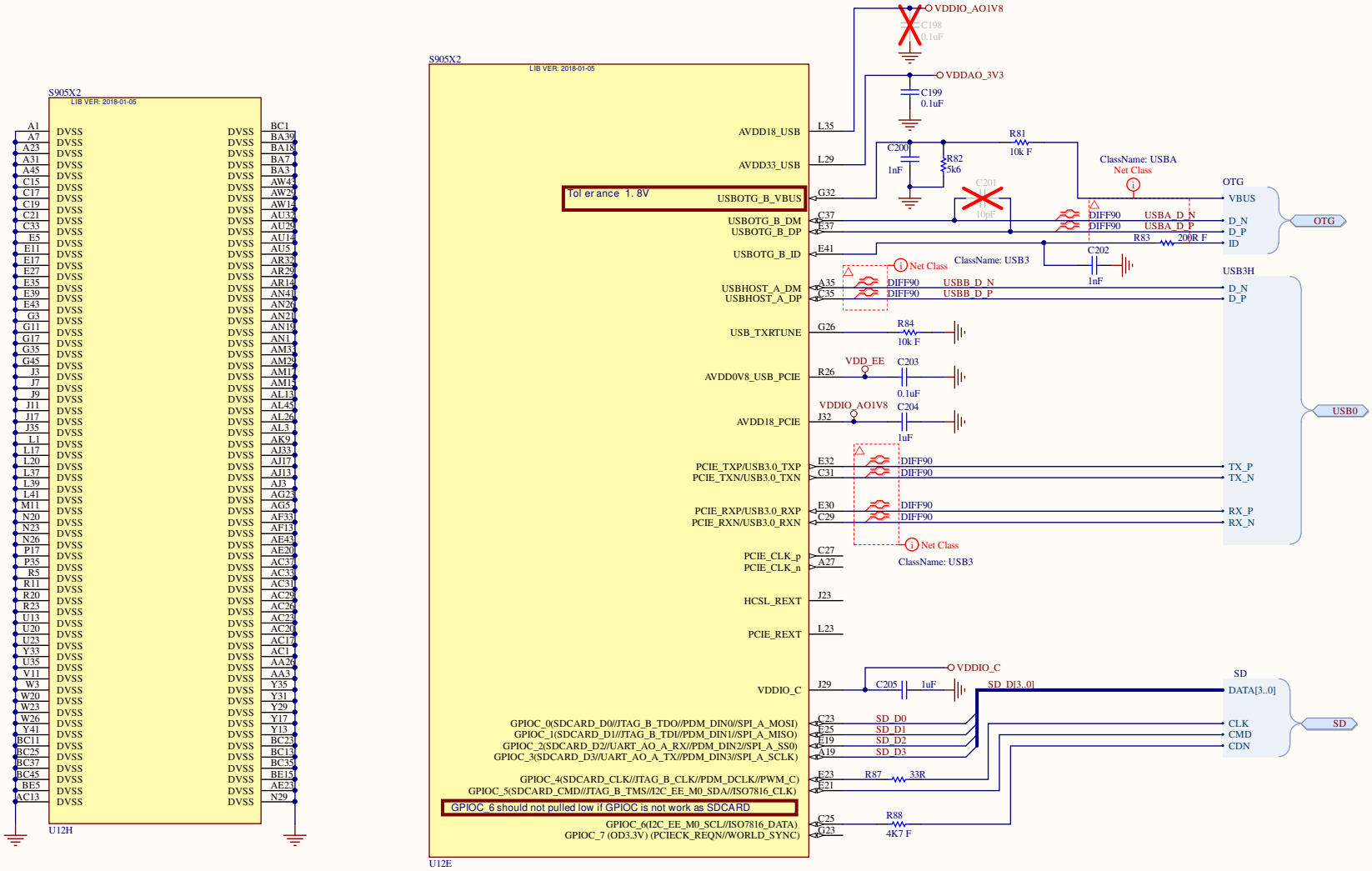


Title S905X2_FLASH			Hardkernel Co., LTD. 475-1 Mananro Manangu, Anyang City Kyeonggido South Korea
Size: A4	Number: 7	Revision: 0.4	
Date: 2/17/2020 Time: 2:23:10 PM	Sheet 7 of 33		
File: S905X2_FLASH.SchDoc		Designed by: neal.kim@hardkernel.com	

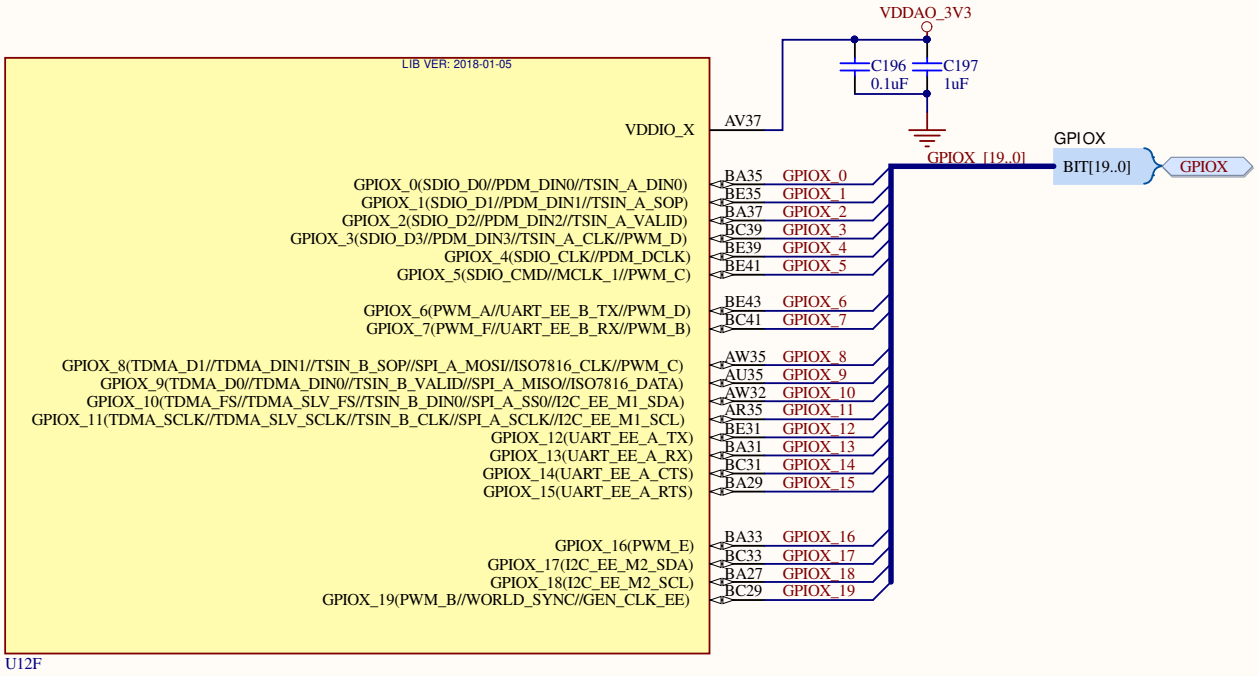


Title	S905X2_HDMI		Hardkernel Co., LTD 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Size: A3	Number: 8	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:10 PM	Sheet 8 of 33	
File: S905X2_HDMI.SchDoc		Designed by: neal.kim@hardkernel.com	

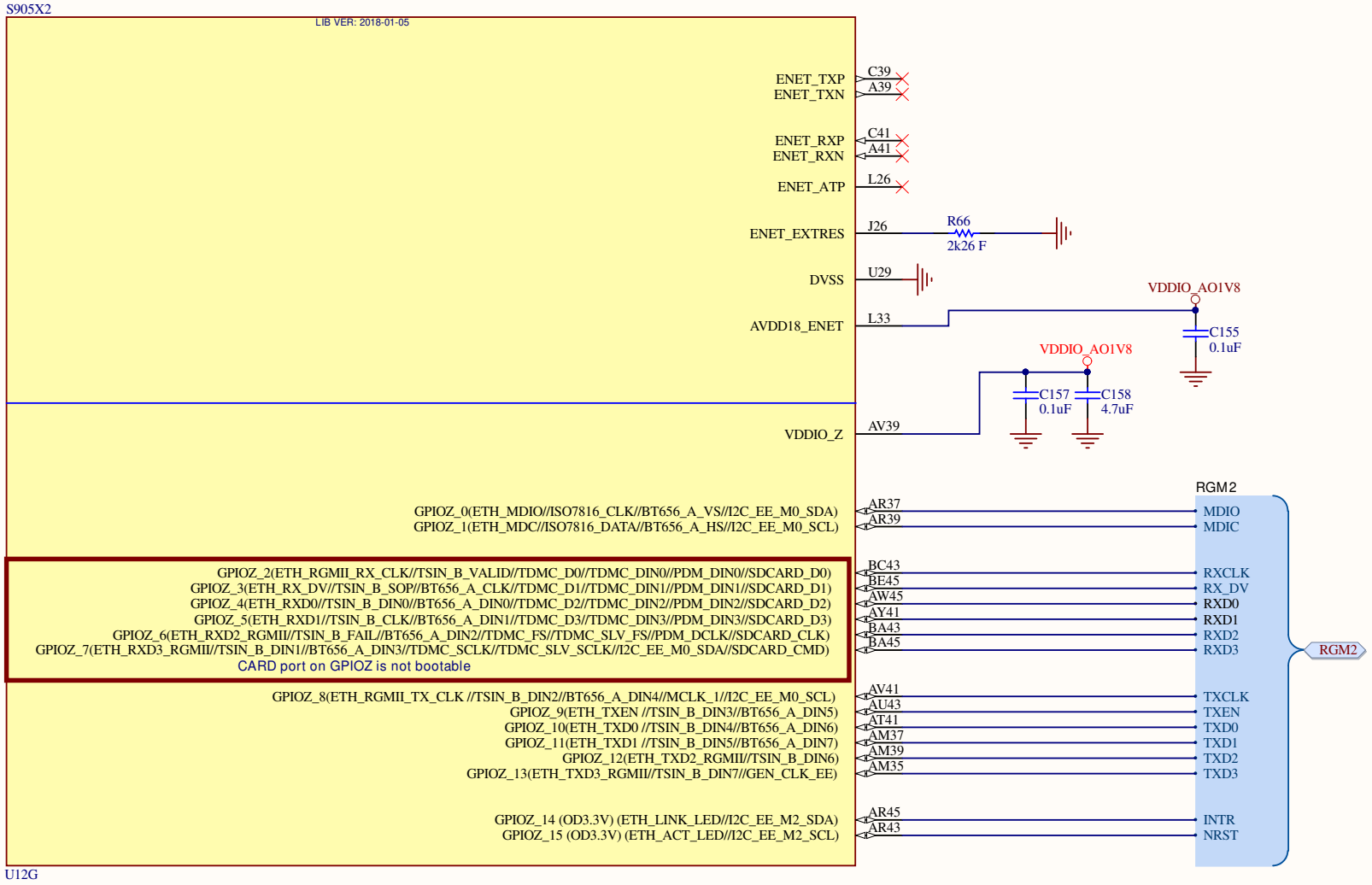
ODROID



Title	S905X2_USB_TFLASH		Hardkernel Co., LTD 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Size: A3	Number: 9	Revision: 0.4	
Date: 2/17/2020 Time: 2:23:11 PM	Sheet 9 of 33		
File: S905X2_USB_TFLASH.SchDoc		Designed by: neal.kim@hardkernel.com	



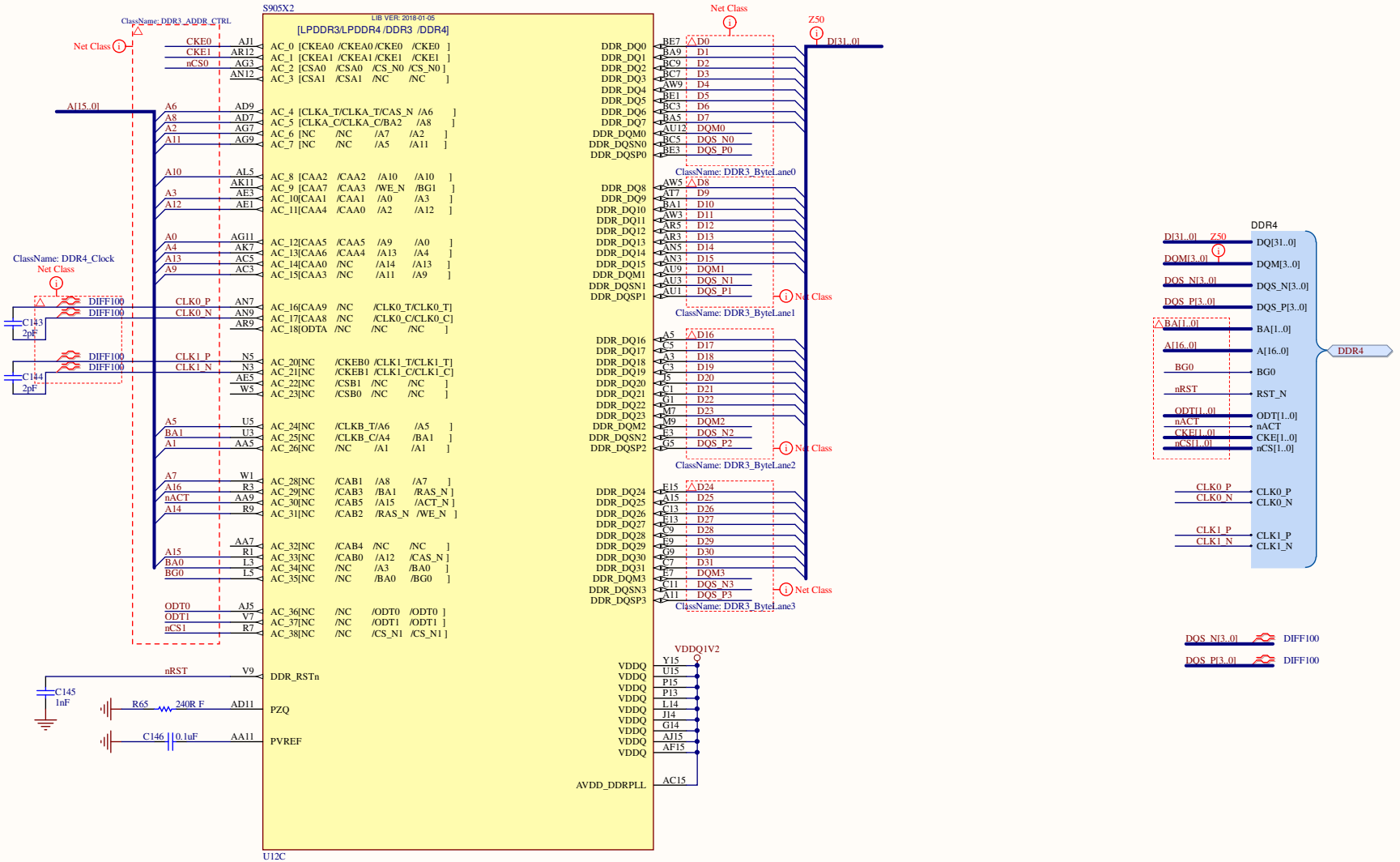
Title S905X2_SDIO			Hardkernel Co., LTD 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Size: A4	Number: 10	Revision: 0.4	
Date: 2/17/2020 Time: 2:23:11 PM	Sheet 10 of 33		
File: S905X2_SDIO.SchDoc		Designed by: neal.kim@hardkernel.com	



Title	S905X2_ETH		
Size: A4	Number: 11	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:11 PM	Sheet 11 of 33	
File: S905X2_ETH.SchDoc			

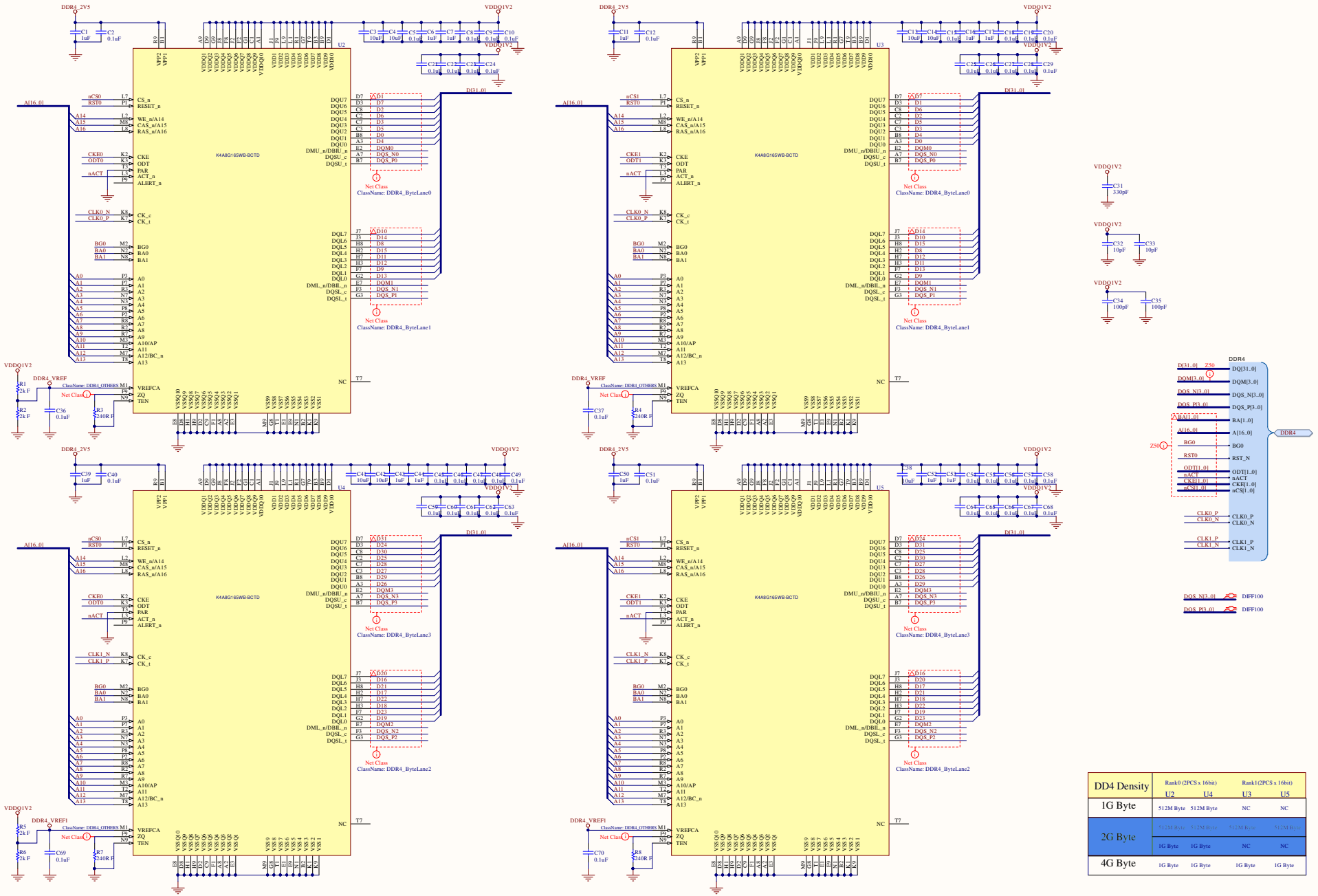
Hardkernel Co., LTD
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea
Designed by: neal.kim@hardkernel.com

ODROID



Title	S905X2_DDR4		Hardkernel Co., LTD 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Size: A3	Number: 12	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:11 PM	Sheet 12 of 33	
File: S905X2_DDR4.SchDoc		Designed by: neal.kim@hardkernel.com	

ODROID

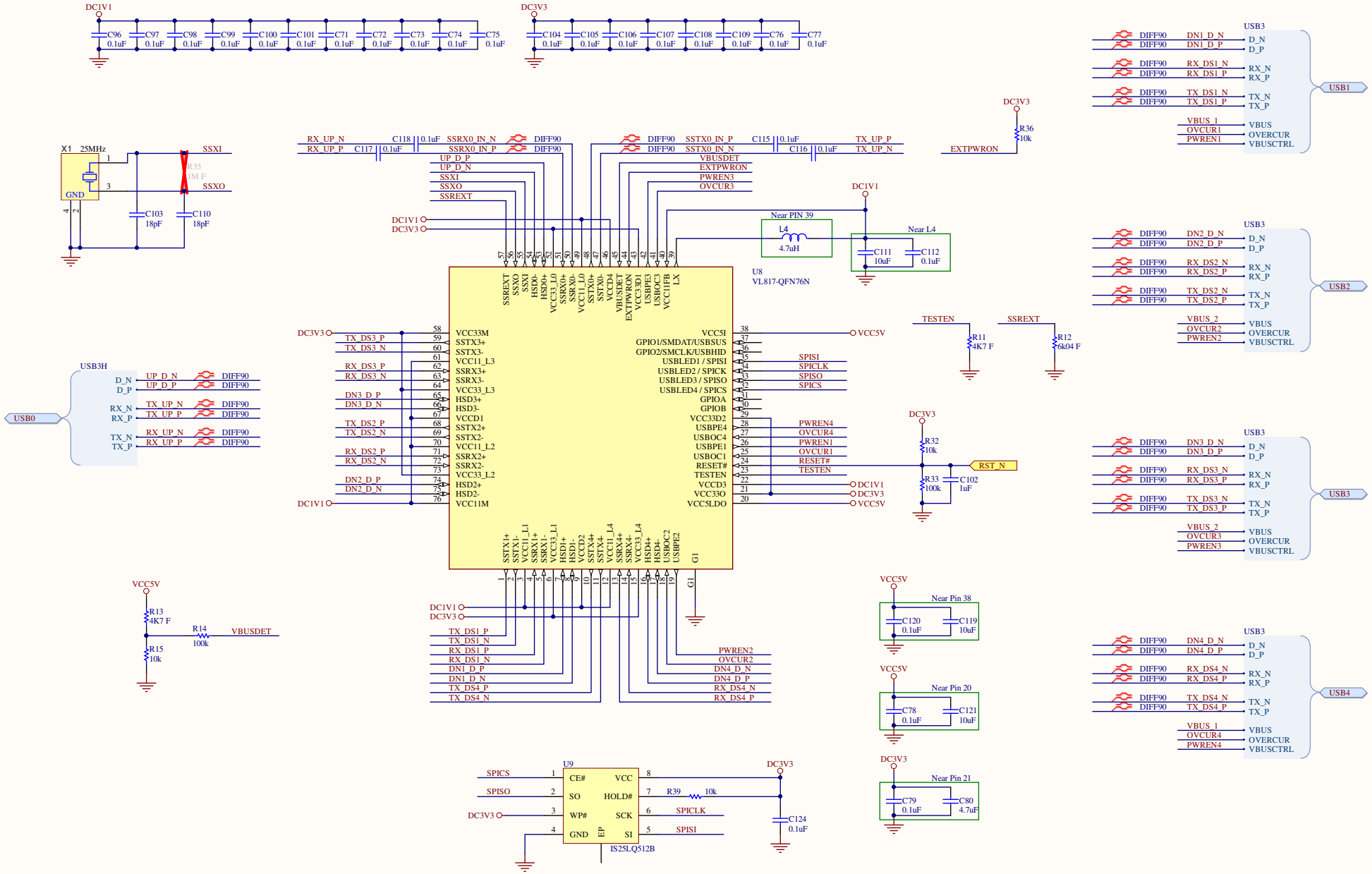


DD4 Density	Rank0 (2PCS x 16bit)		Rank1(2PCS x 16bit)	
	U2	U4	U3	U5
1G Byte	512M Byte	512M Byte	NC	NC
2G Byte	512M Byte	512M Byte	512M Byte	512M Byte
	1G Byte	1G Byte	NC	NC
4G Byte	1G Byte	1G Byte	1G Byte	1G Byte

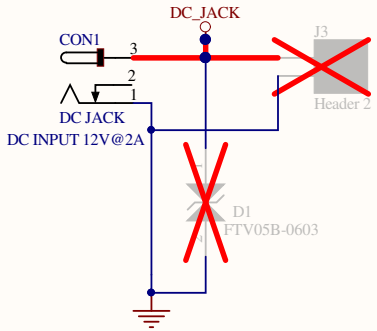
Title	DDR4 16b 8G x 4		
Size: A2	Number: 13	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:11 PM	Sheet 13 of 33	
File: MEM_DDR4_8GbX4.SchDoc			

Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea
Designed by: nuppi.kim@hardkernel.com

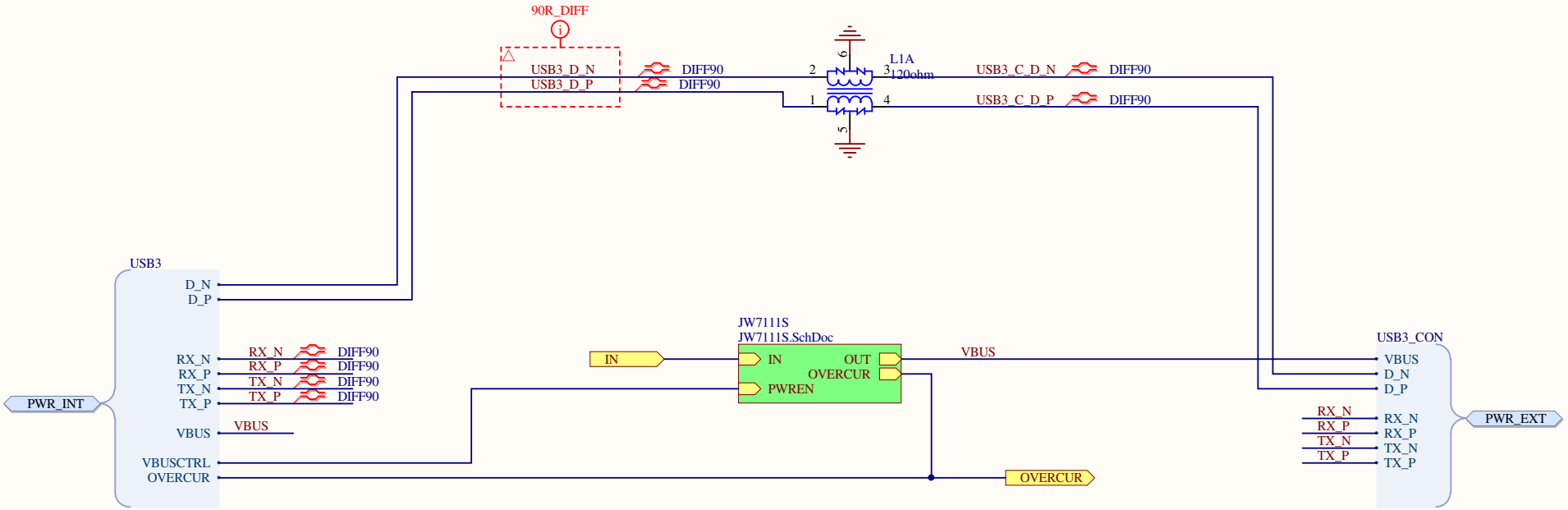
ODROID



Title	USB HUB VL817Q7		Hardkernel Co.,LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Size: A3	Number: 14	Revision: 0.4	ODROID
Date: 2/17/2020	Time: 2:23:12 PM	Sheet 14 of 33	
File: USB_HUB_VL817Q7.SchDoc		Designed by: neal.kim@hardkernel.com	



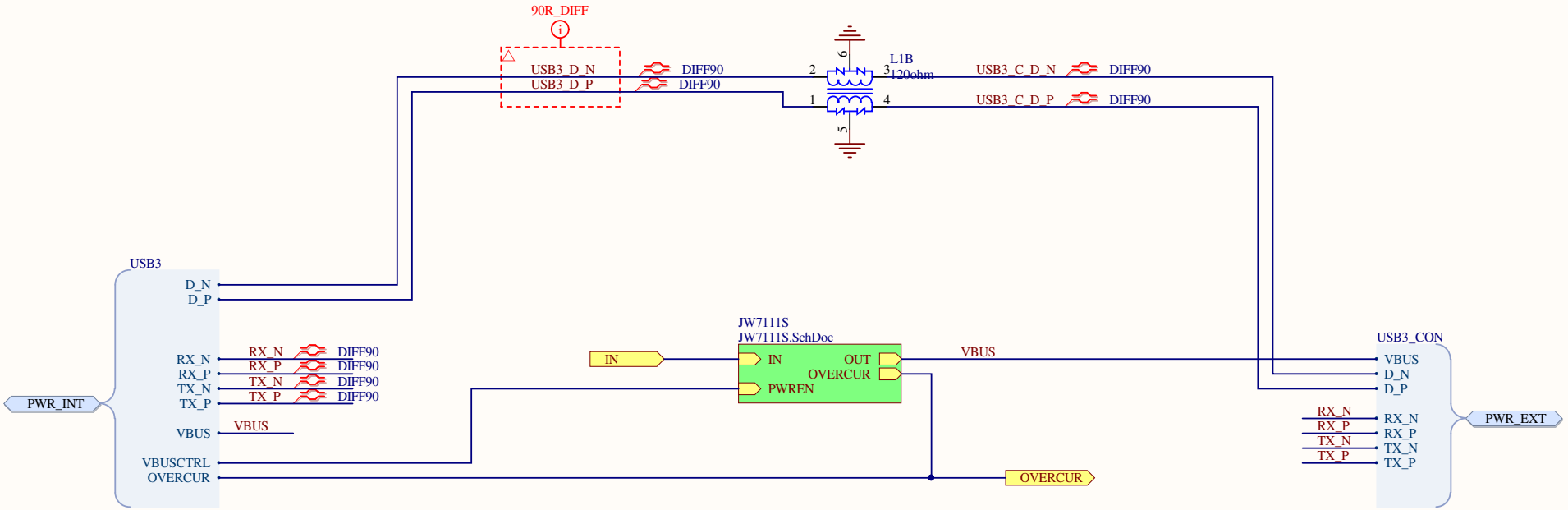
Title DC JACK			Hardkernel Co.,LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Size: A4	Number: 15	Revision: 0.4	ODROID
Date: 2/17/2020 Time: 2:23:13 PM	Sheet 15 of 33		
File: CON_DCJACK.SchDoc		Designed by: ruppi.kim@hardkernel.com	



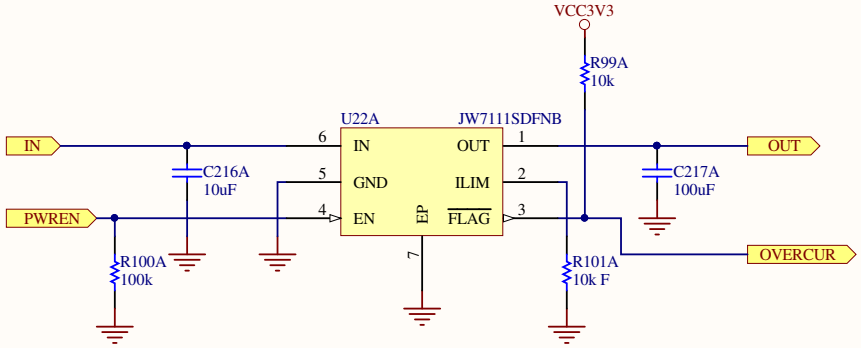
Title	USB3 BUS Power Switch		
Size: A4	Number: 16	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:13 PM	Sheet 16 of 33	
File: USB3_BUS_PWR.SchDoc			

Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea
Designed by: ruppi kim@hardkernel.com

ODROID



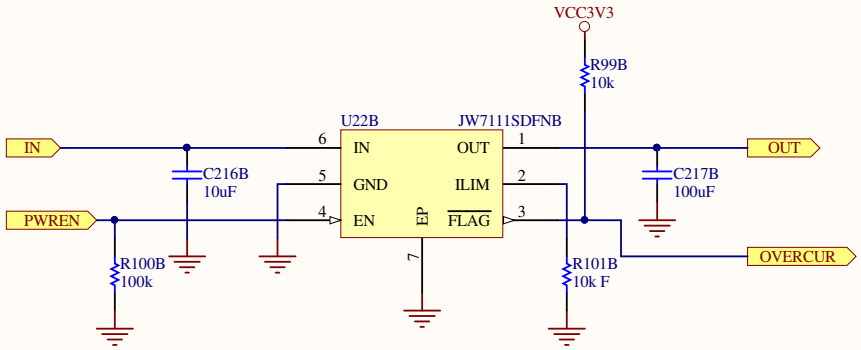
Title	USB3 BUS Power Switch		Hardkernel Co.,LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Size: A4	Number: 16	Revision: 0.4	
Date: 2/17/2020 Time: 2:23:13 PM	Sheet 16 of 33		
File: USB3_BUS_PWR.SchDoc		Designed by: ruppi kim@hardkernel.com	



Title	USB HOST Power Switch		
Size: A4	Number: 17	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:13 PM	Sheet 17 of 33	
File: JW7111S.SchDoc			

Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea
Designed by: ruppi kim@hardkernel.com

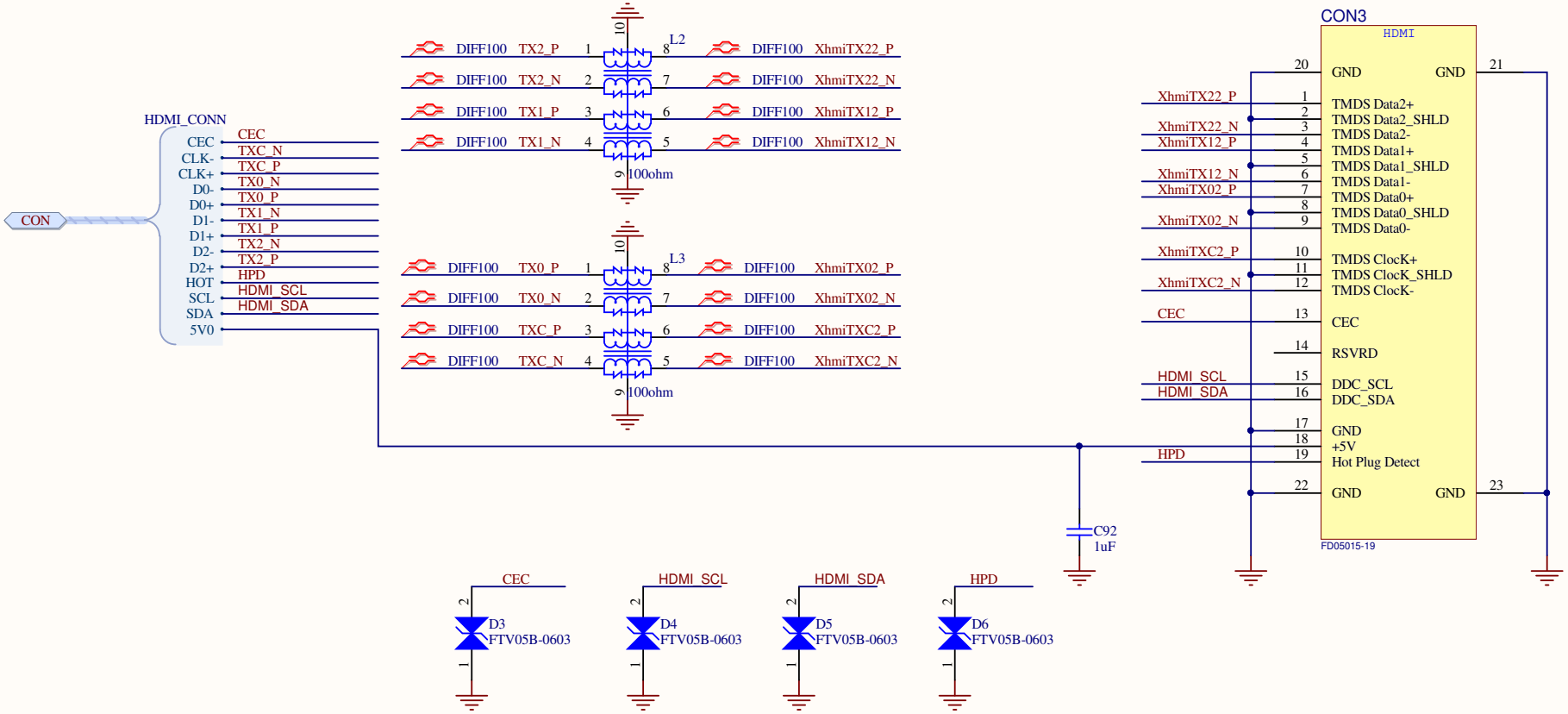
ODROID



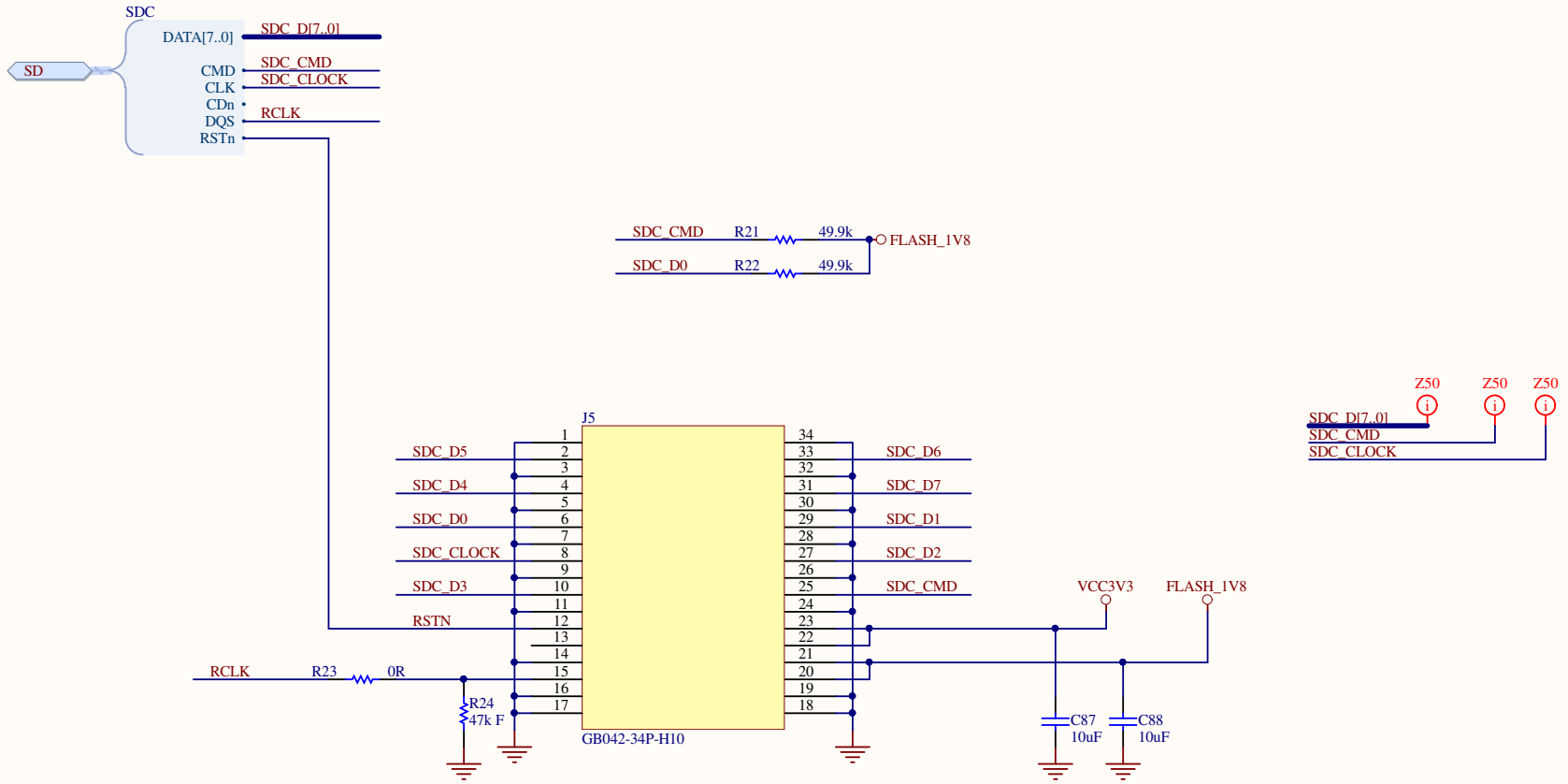
Title	USB HOST Power Switch		
Size: A4	Number: 17	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:13 PM	Sheet 17 of 33	
File: JW7111S.SchDoc			

Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea
Designed by: ruppi kim@hardkernel.com

ODROID



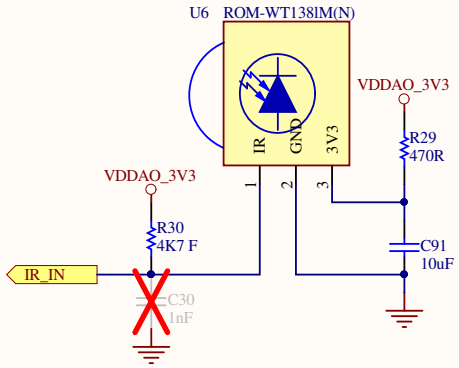
Title: HDMI Connector			Hardkernel Co.,LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Size: A4	Number: 18	Revision: 0.4	
Date: 2/17/2020 Time: 2:23:13 PM	Sheet 18 of 33		
File: CON_HDMI.SchDoc		Designed by: ruppi.kim@hardkernel.com	



Title	eMMC B2B Connector		
Size: A4	Number: 19	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:13 PM	Sheet 19 of 33	
File: CON_EMMC.SchDoc			

Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea
Designed by: ruppi.kim@hardkernel.com

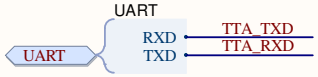
ODROID

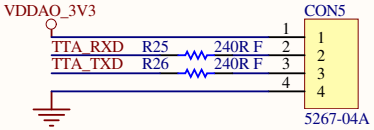


Title	Remote Controller Receiver		
Size: A4	Number: 20	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:13 PM	Sheet 20 of 33	
File: IR.SchDoc			

Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea
Designed by: ruppi.kim@hardkernel.com

ODROID

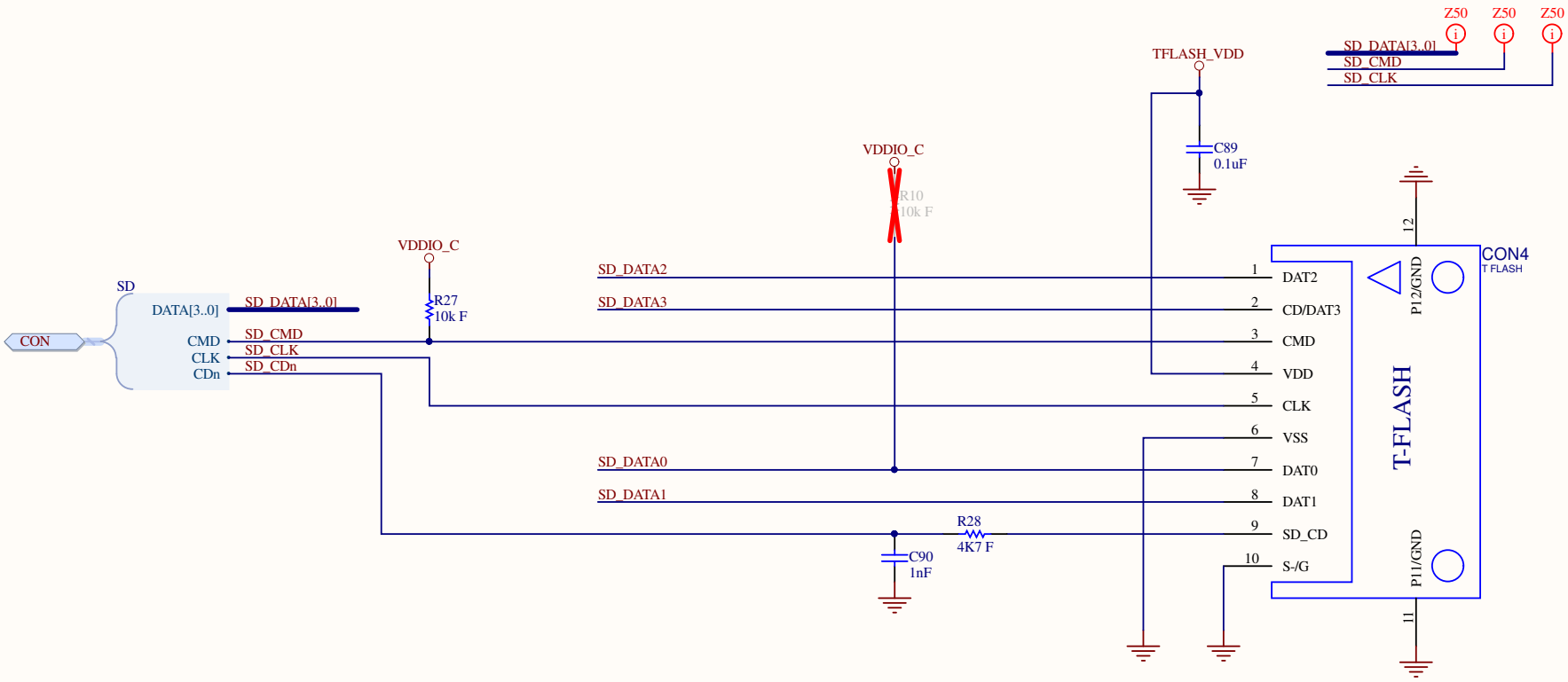




Title	UART Connector		
Size: A4	Number: 21	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:13 PM	Sheet 21 of 33	
File: CON_UART.SchDoc			

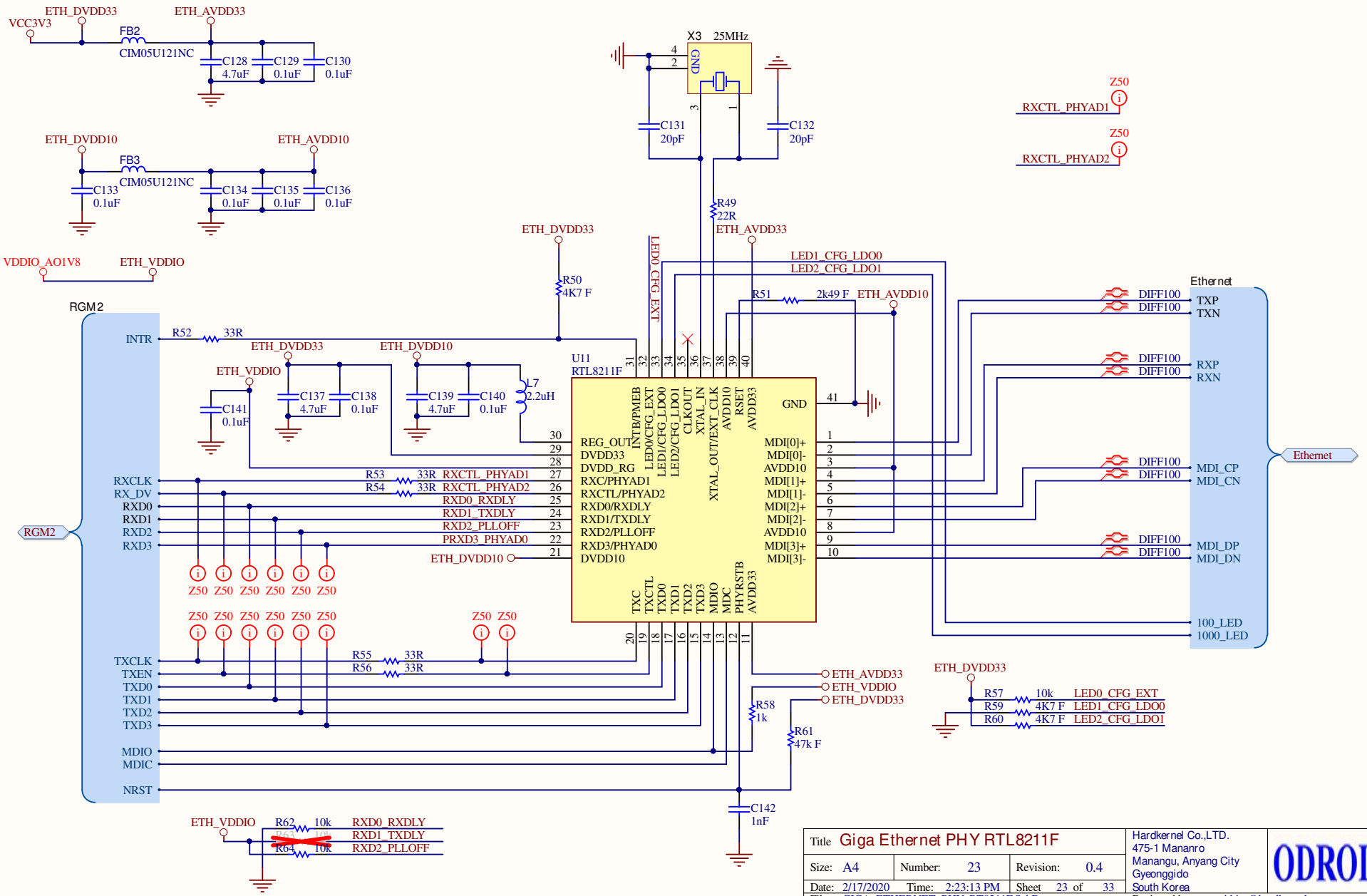
Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea
Designed by: ruppi.kim@hardkernel.com

ODROID



Title	MicroSD Socket		
Size: A4	Number: 22	Revision: 0.4	Hardkernel Co.,LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Date: 2/17/2020	Time: 2:23:13 PM	Sheet 22 of 33	
File: CON_MICROSD_HD08-AL00335.SchDoc		Designed by: ruppi.kim@hardkernel.com	

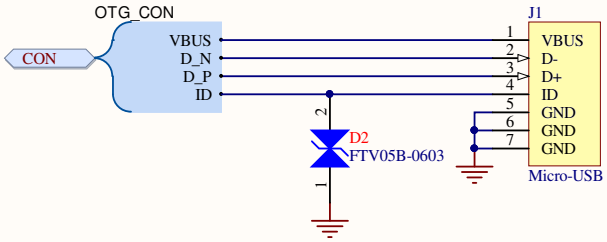
ODROID



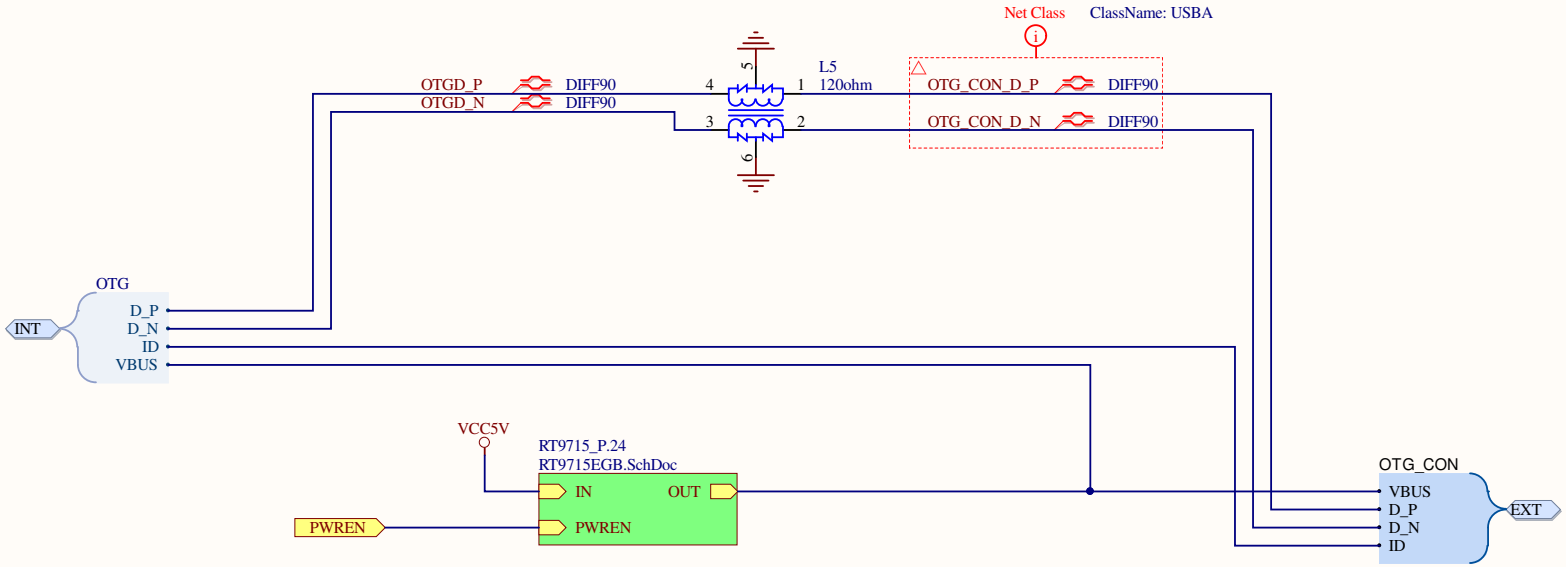
Title	Giga Ethernet PHY RTL8211F		
Size: A4	Number: 23	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:13 PM	Sheet 23 of 33	
File: GIGA_ETHERNET_PHY_RT8211F.SchDoc		Designed by: ruppi.kim@hardkernel.com	

Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea

ODROID



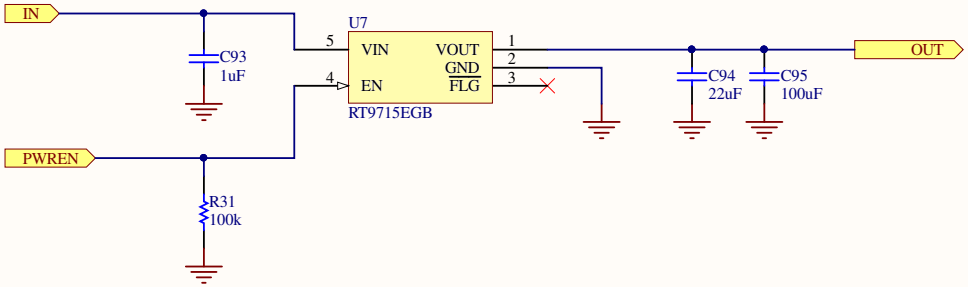
Title: Micro USB Connector			Hardkernel Co.,LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Size: A4	Number: 24	Revision: 0.4	
Date: 2/17/2020 Time: 2:23:13 PM	Sheet 24 of 33		
File: CON_MICROUSB.SchDoc		Designed by: ruppi.kim@hardkernel.com	



Title	USB OTG Power Switch		
Size: A4	Number: 25	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:14 PM	Sheet 25 of 33	
File: USB_OTG.SchDoc			

Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea
Designed by: ruppi.kim@hardkernel.com

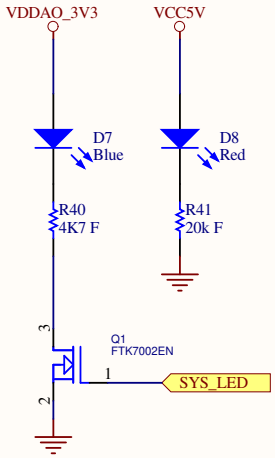
ODROID



Title	USB HOST PWR RT9715		
Size: A4	Number: 26	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:14 PM	Sheet 26 of 33	
File: RT9715EGB.SchDoc			

Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea
Designed by: ruppi.kim@hardkernel.com

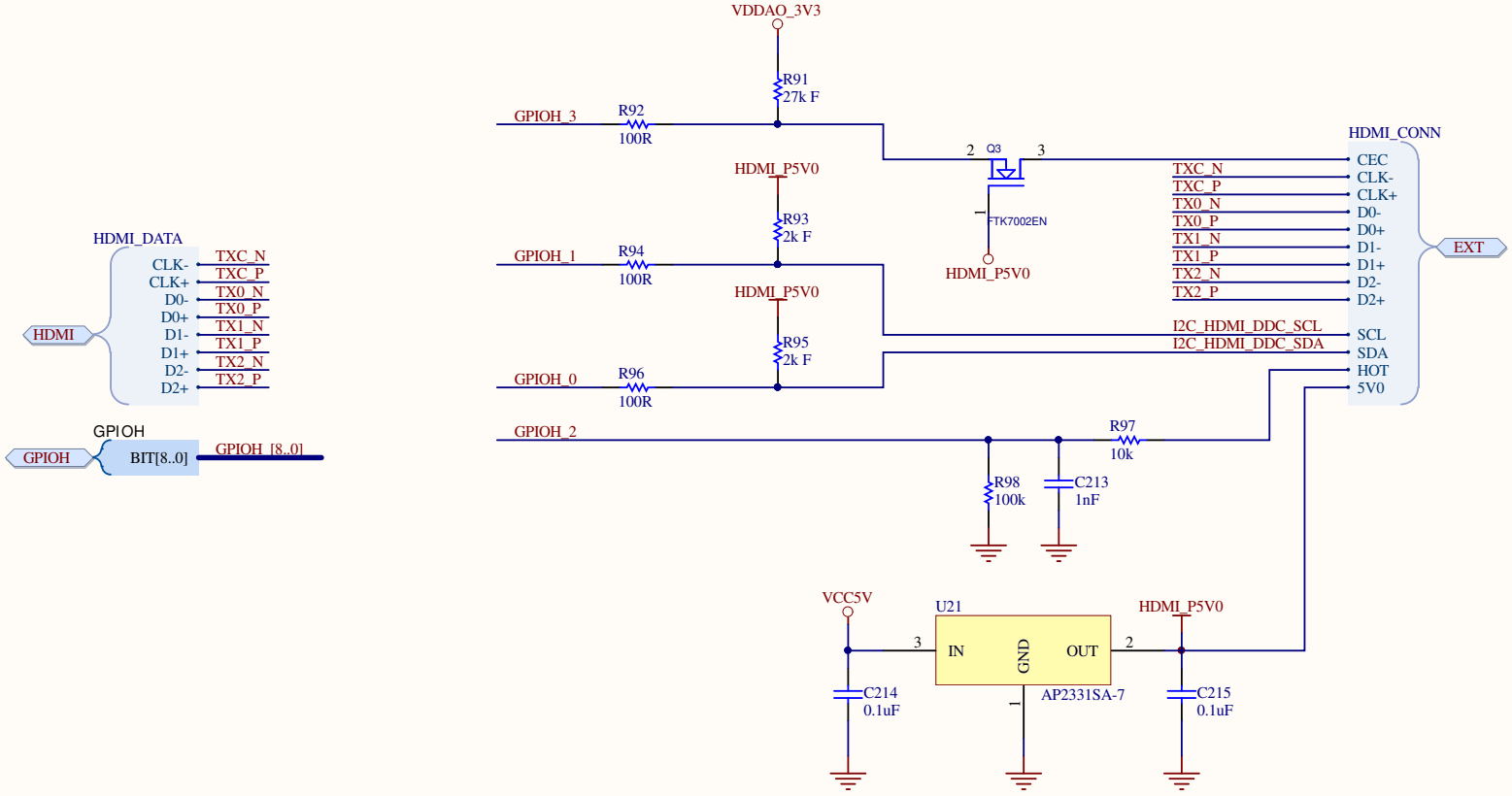
ODROID



Title	PWR SYSTEM LEDs		
Size: A4	Number: 27	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:14 PM	Sheet 27 of 33	
File: C_LEDS.SchDoc			

Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea
Designed by: *@hardkernel.com

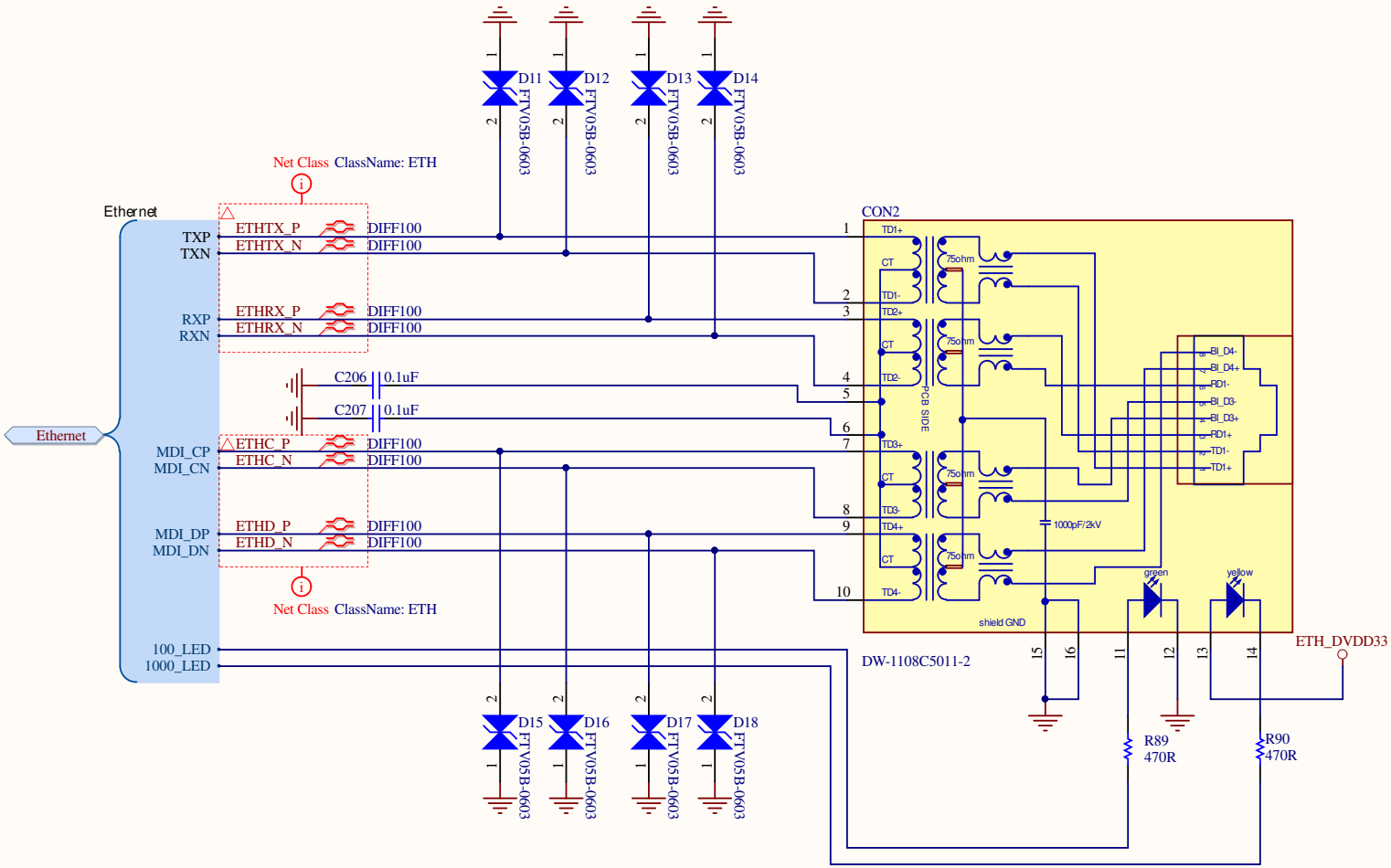
ODROID



Title	HDMI I2C		
Size: A4	Number: 28	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:14 PM	Sheet 28 of 33	
File: HDMI_I2C.SchDoc			

Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea
Designed by: ruppi.kim@hardkernel.com

ODROID

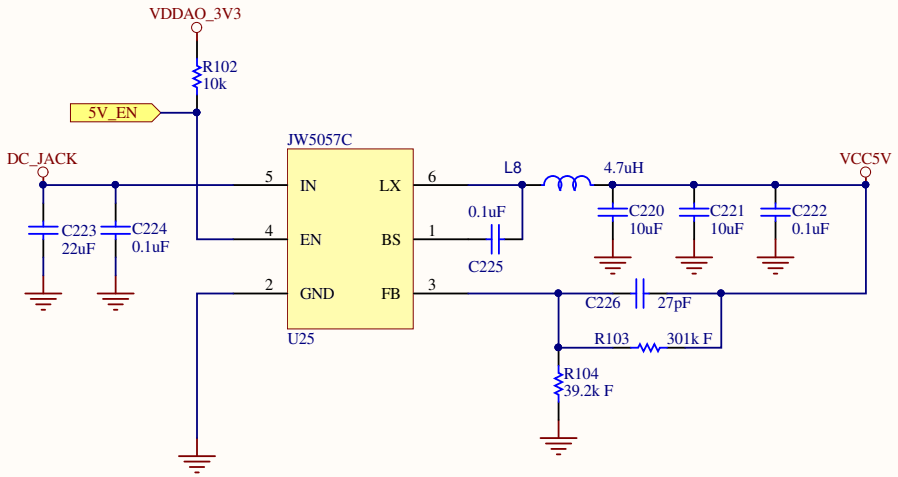


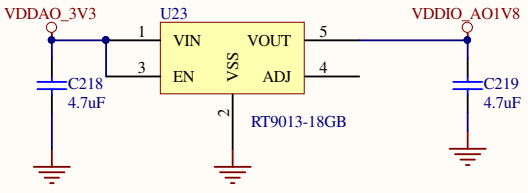
Title	Giga Ethernet Connector		
Size: A4	Number: 29	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:14 PM	Sheet 29 of 33	
File: CON_GIGA_ETH.SchDoc			

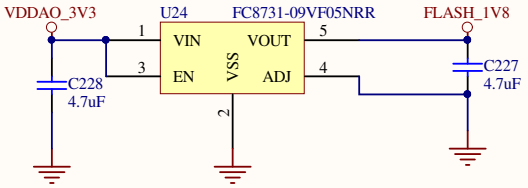
Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea

Designed by: ruppi.kim@hardkernel.com

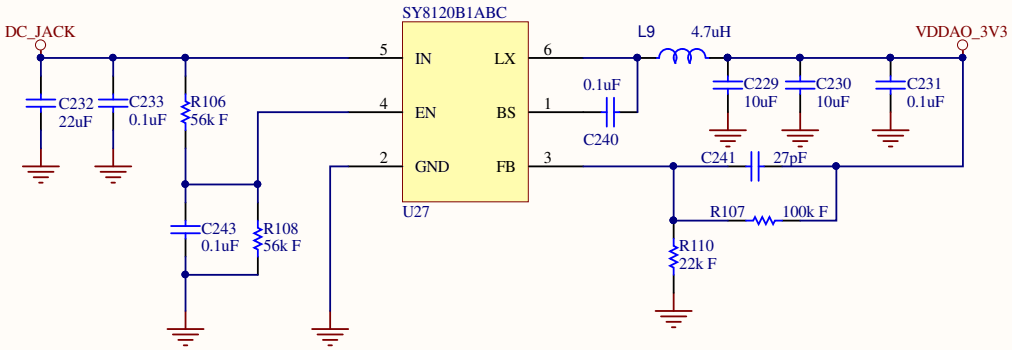
ODROID

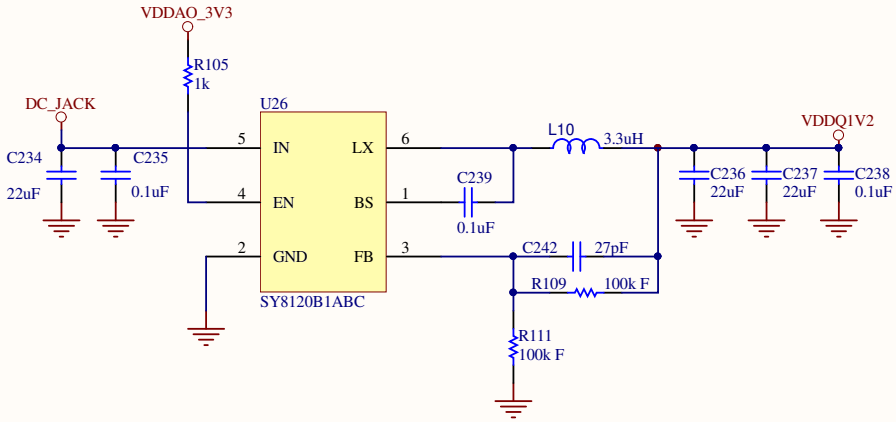








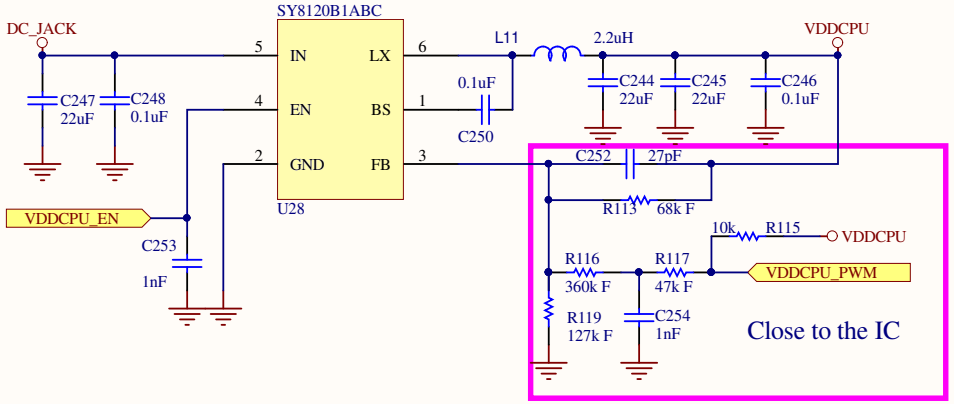


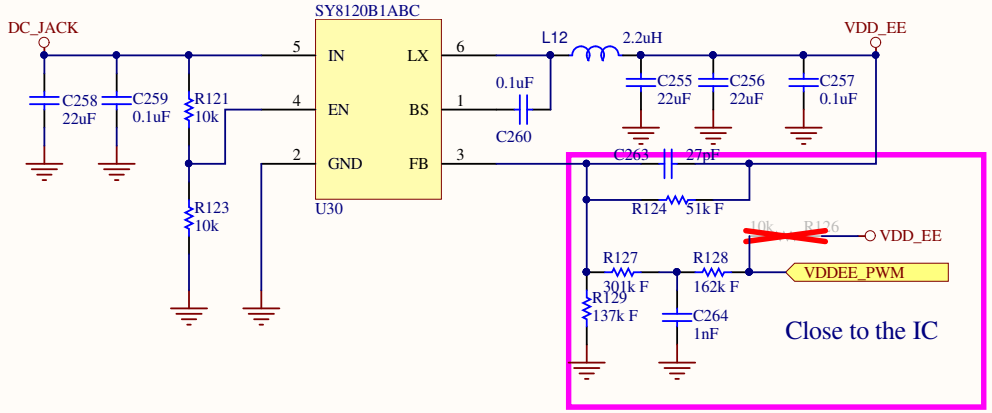


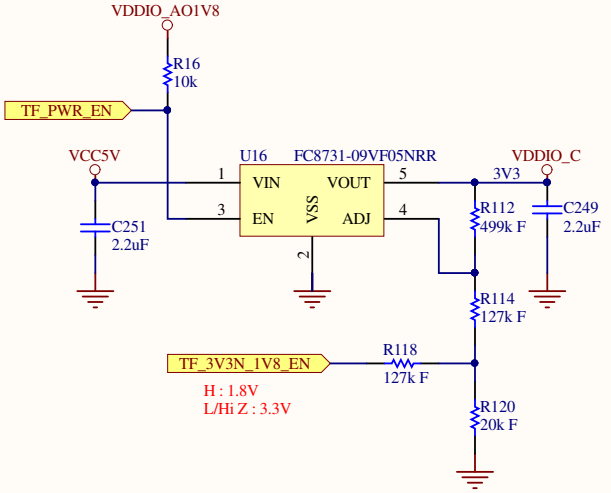
Title	PWR_DCDC_5V_3V3		
Size: A4	Number: 30	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:14 PM	Sheet 30 of 33	
File: PWR_DCDC_5V_3V3.SchDoc		Designed by: neal.kim@hardkernel.com	

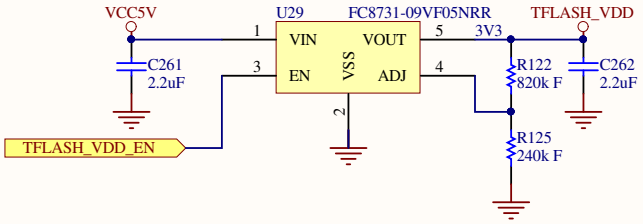
Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea

ODROID

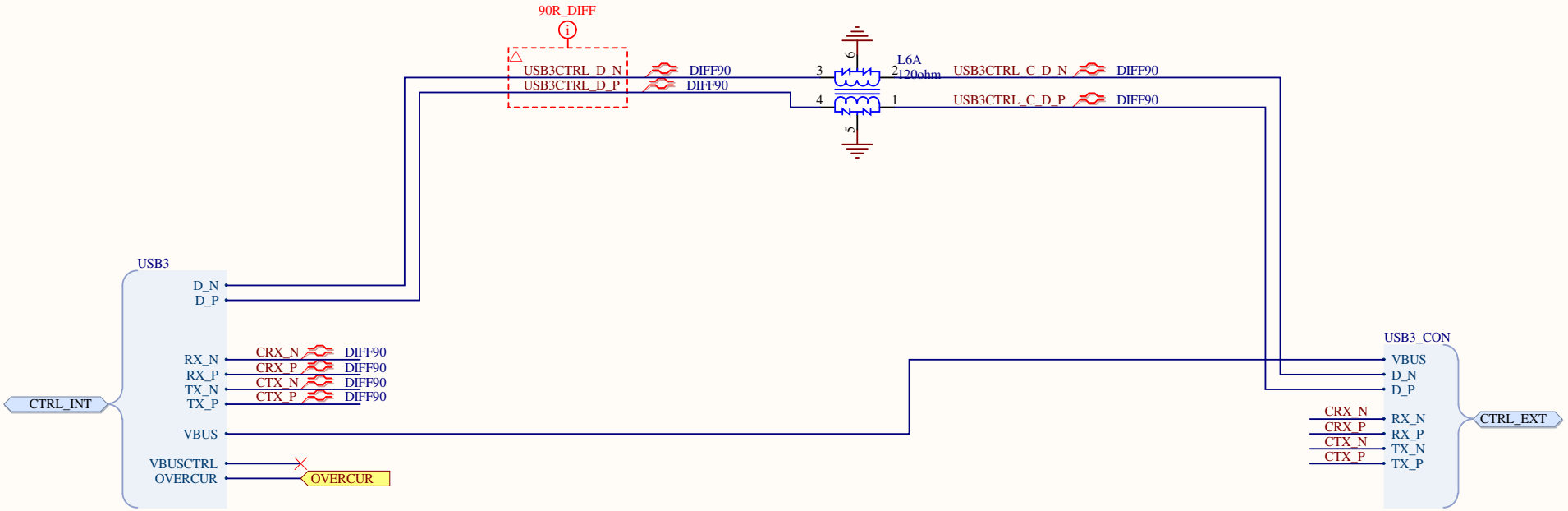








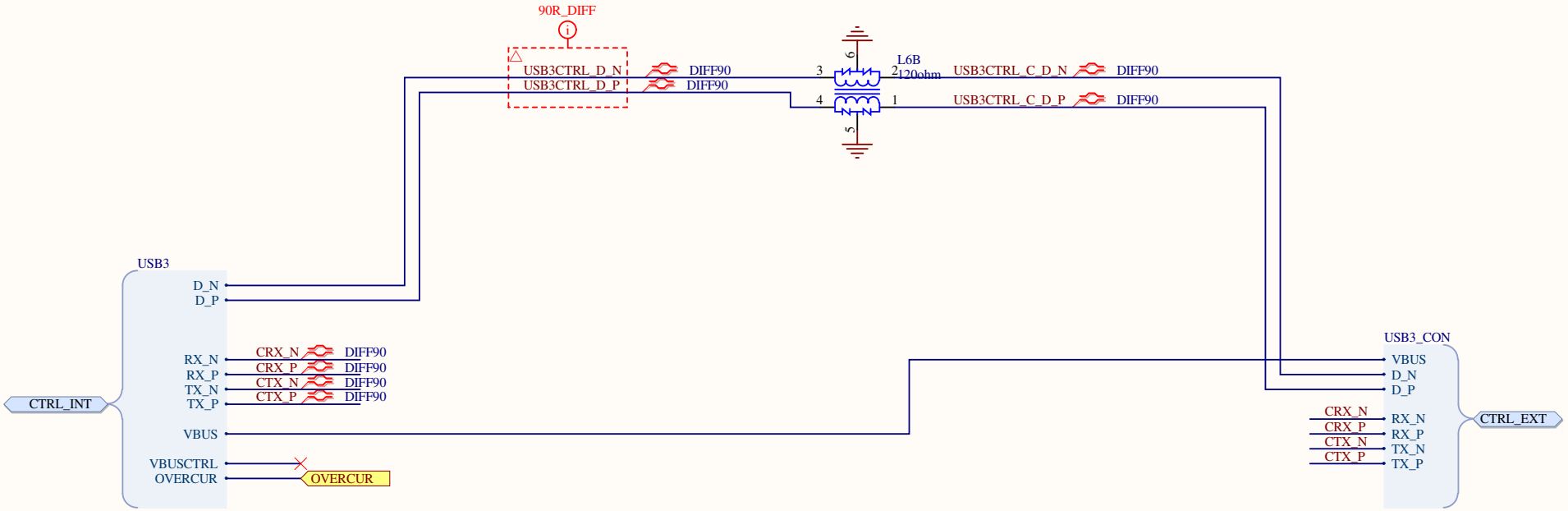
Title PWR_DCDC_OTHERS			Hardkernel Co.,LTD. 475-1 Mananro Manangu, Anyang City Gyeonggido South Korea
Size: A4	Number: 31	Revision: 0.4	
Date: 2/17/2020 Time: 2:23:14 PM	Sheet 31 of 33		
File: PWR_DCDC_OTHERS.SchDoc		Designed by: neal.kim@hardkernel.com	



Title	USB3_BUS_CTRL		
Size: A4	Number: 32	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:14 PM	Sheet 32 of 33	
File: USB3_BUS_CTRL.SchDoc			

Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea
Designed by: neal.kim@hardkernel.com

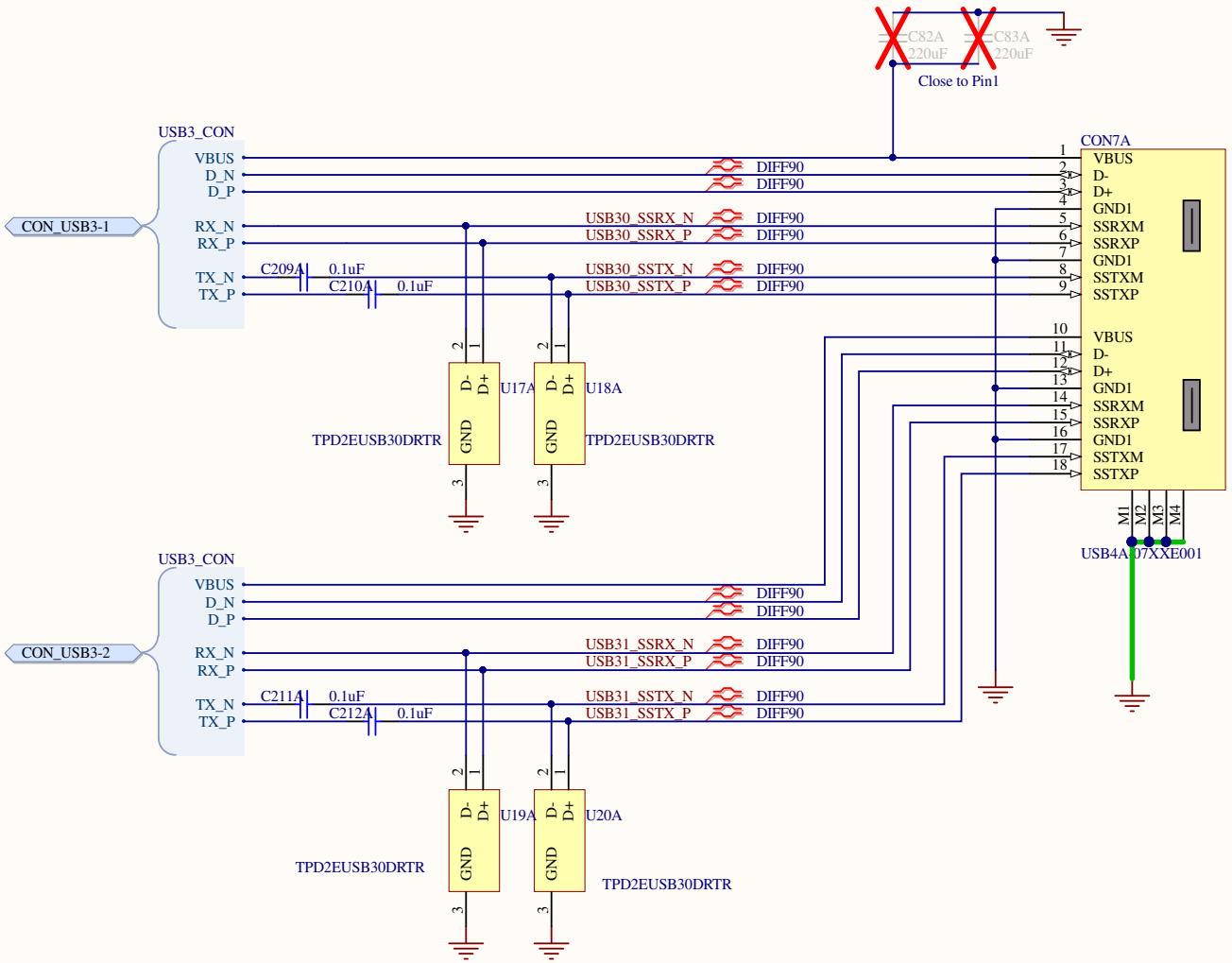
ODROID



Title	USB3_BUS_CTRL		
Size: A4	Number: 32	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:14 PM	Sheet 32 of 33	
File: USB3_BUS_CTRL.SchDoc			

Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea
Designed by: neal.kim@hardkernel.com

ODROID

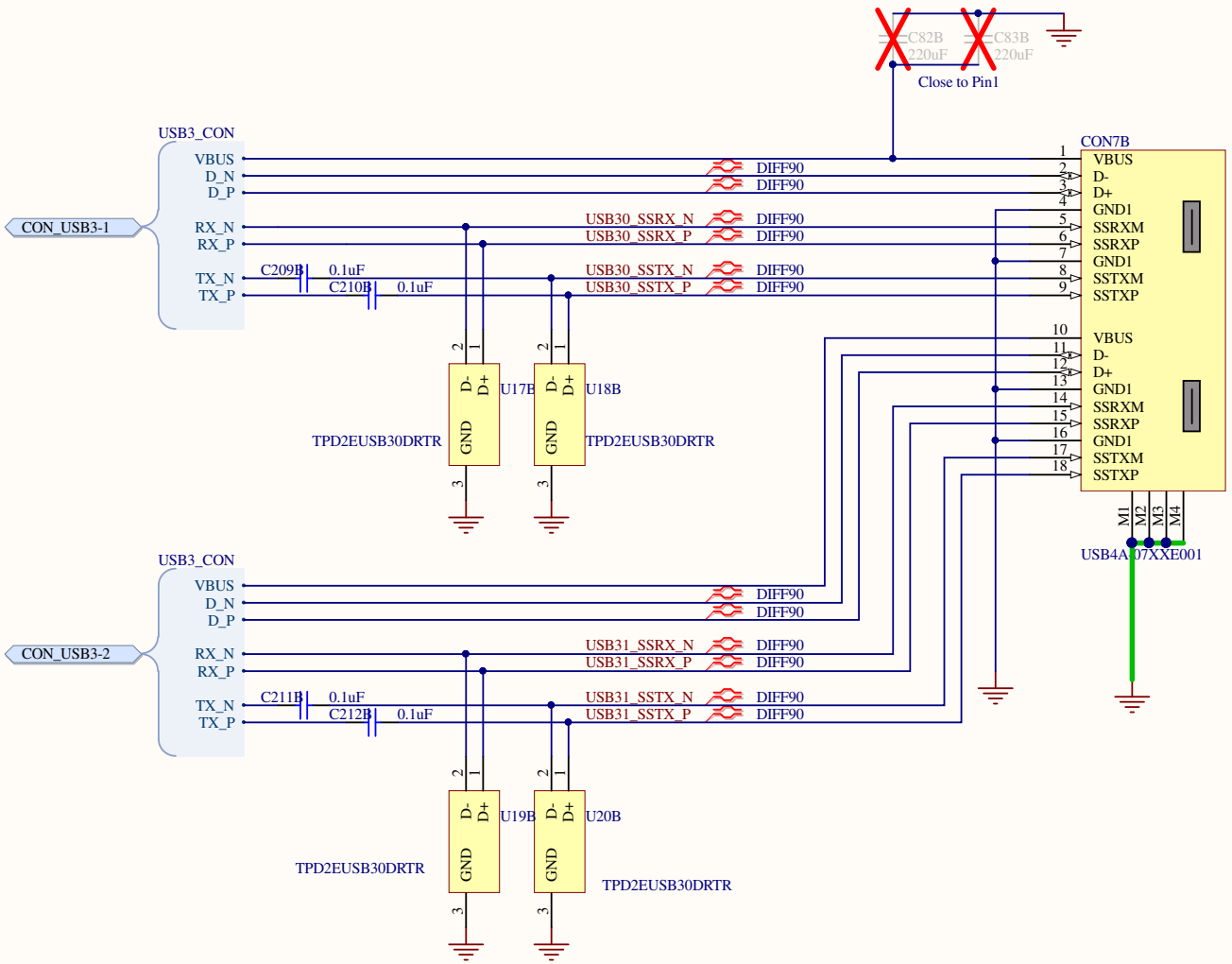


Title	USB 3.0 Type A Connector		
Size: A4	Number: 33	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:14 PM	Sheet 33 of 33	
File: CON_USB3_DUAL_RA.SchDoc			

Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea

Designed by: ruppi kim@hardkernel.com

ODROID



Title	USB 3.0 Type A Connector		
Size: A4	Number: 33	Revision: 0.4	
Date: 2/17/2020	Time: 2:23:14 PM	Sheet 33 of 33	
File: CON_USB3_DUAL_RA.SchDoc			

Hardkernel Co.,LTD.
475-1 Mananro
Manangu, Anyang City
Gyeonggido
South Korea
Designed by: ruppi kim@hardkernel.com

ODROID